

Large Language Models

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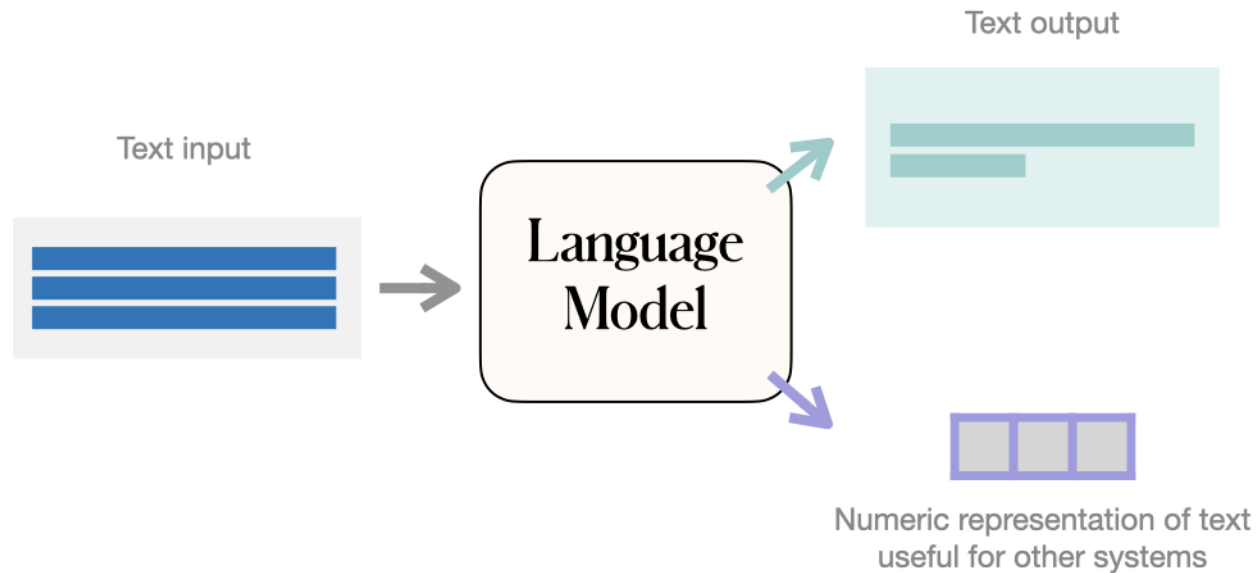
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Lecture goal

- Language Models: recap of fundamentals
- Autoregressive LMs and conditional text generation
- Key LLM features and categories
- Main LLM architectures and pretraining objectives
- LLM adaptation
- LLM applications
- LLM-based agents

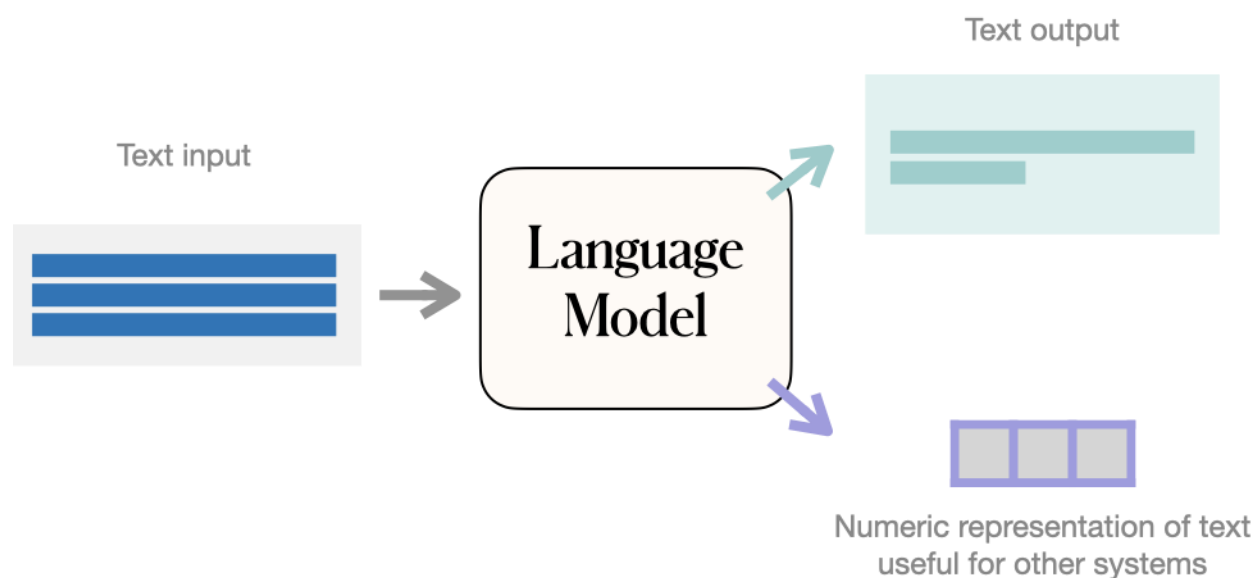
Language Model

- Model the generative likelihood of word sequences to predict the probabilities of future (or missing) tokens



Language Model

- Given a sequence, the LM returns a probability to assess its goodness
- Probability distribution over sequences of tokens in a vocabulary
 - The probability indicates how good that sequence of tokens is



Language Models: fundamentals

- Predict the probabilities of future (or missing) tokens
- Training sequence of tokens: $y_1, \dots, y_n$
- Model prediction (timestep t): $p^{(t)} = p(*|y_1, \dots, y_{t-1})$
- Target (timestep t): $p^* = \text{one-hot}(y_t)$
- Loss function: $Loss(p^*, p) = -p^* \log(p) = -\sum_{i=1}^{|V|} p_i^* \log(p_i)$

Language Models: fundamentals

- Since just one p^* term is non-zero, i.e., the correct token y_t

$$Loss(p^*, p) = -\log(p_{y_t}) = -\log(p(y_t|y_{<t}))$$

- For the whole sequence:

$$-\sum_{t=1}^n \log(p(y_t|y_{<t}))$$

Autoregressive Language Models

- Conditional probability distribution $p(x_i \mid x_{1:i-1})$
 - Can be estimated by an autoregressive model like feedforward neural network
 - Chain rule of probability:

$$p(x_{1:L}) = p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_1, x_2) \cdots p(x_L \mid x_{1:L-1}) = \prod_{i=1}^L p(x_i \mid x_{1:i-1})$$

- E.g.,

$$\begin{aligned} p(\text{the, mouse, ate, the, cheese}) &= p(\text{the}) \\ &\quad p(\text{mouse} \mid \text{the}) \\ &\quad p(\text{ate} \mid \text{the, mouse}) \\ &\quad p(\text{the} \mid \text{the, mouse, ate}) \\ &\quad p(\text{cheese} \mid \text{the, mouse, ate, the}). \end{aligned}$$

Autoregressive Language Models

- To generate the entire sequence from an autoregressive LM
 - Sample one token at a time given the tokens generated so far

for $i = 1, \dots, L$:

$$x_i \sim p(x_i \mid x_{1:i-1})^{1/T}$$

Where $T > 0$ is the temperature parameter that controls the randomness we want from the LM

- $T = 0$ deterministically choose the most likely token at each step i
- $T = 1$ sample from the pure language model using a normal distribution
- $T = \infty$ sample from the uniform distribution over the entire vocabulary

Annealed conditional probability distribution

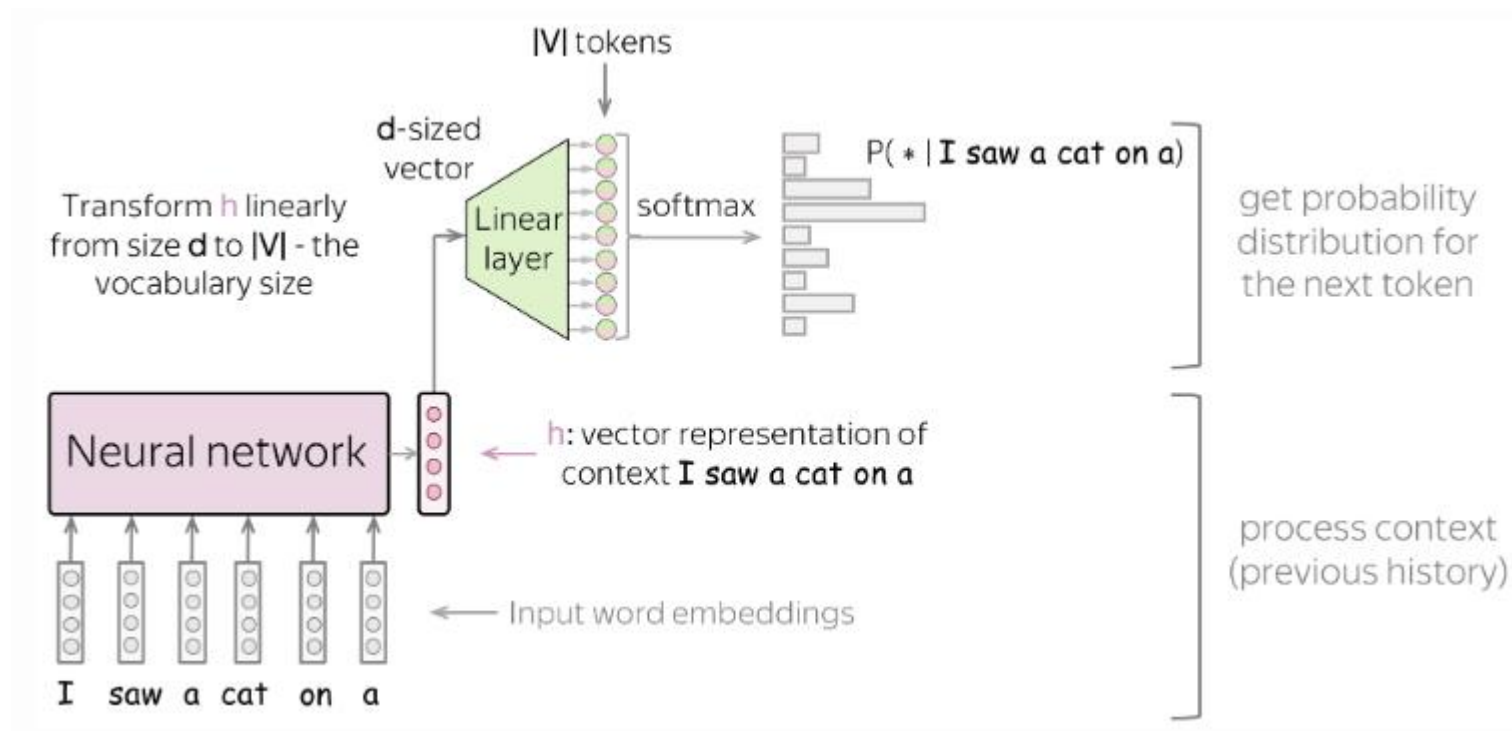
- When T is not equal to 1 the probability distribution need to be re-normalized
 - Otherwise the probabilities do not sum to one

Conditional text generation

- Given a prefix sequence, called **prompt**, generate the rest of the sequence, called **completion**

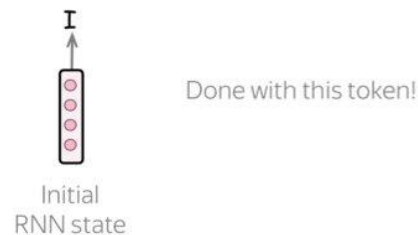
$\underbrace{\text{the, mouse, ate}}_{\text{prompt}} \xrightarrow[T \approx 0]{} \underbrace{\text{the, cheese.}}_{\text{completion}}$

Language Model



Language Model

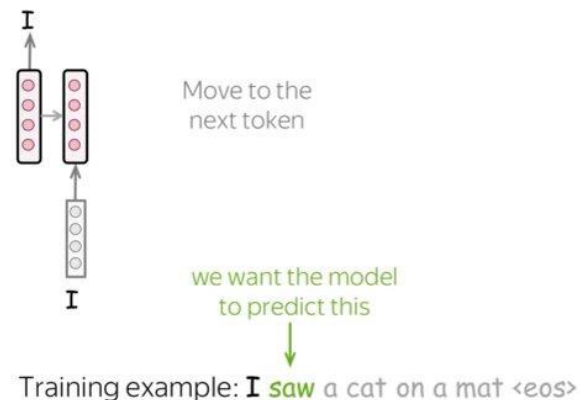
- Model the generative likelihood of word sequences to predict the probabilities of future (or missing) tokens



Training example: **I** saw a cat on a mat <eos>

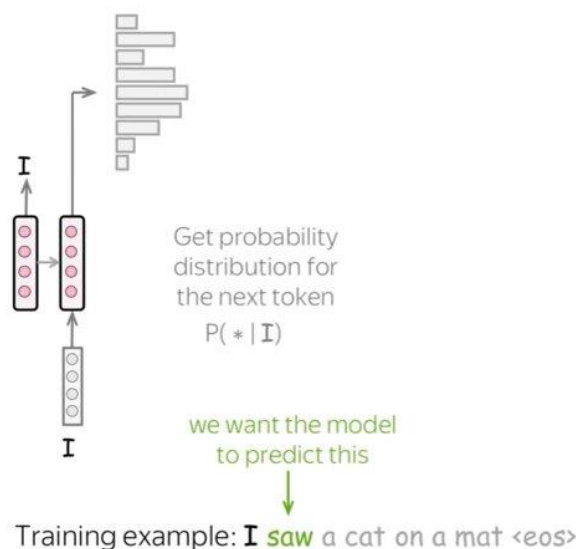
Language Model

- Model the generative likelihood of word sequences to predict the probabilities of future (or missing) tokens



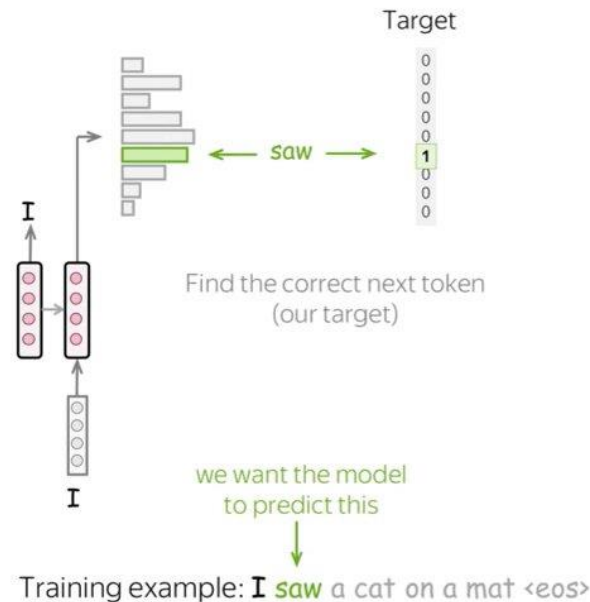
Language Model

- Model the generative likelihood of word sequences to predict the probabilities of future (or missing) tokens



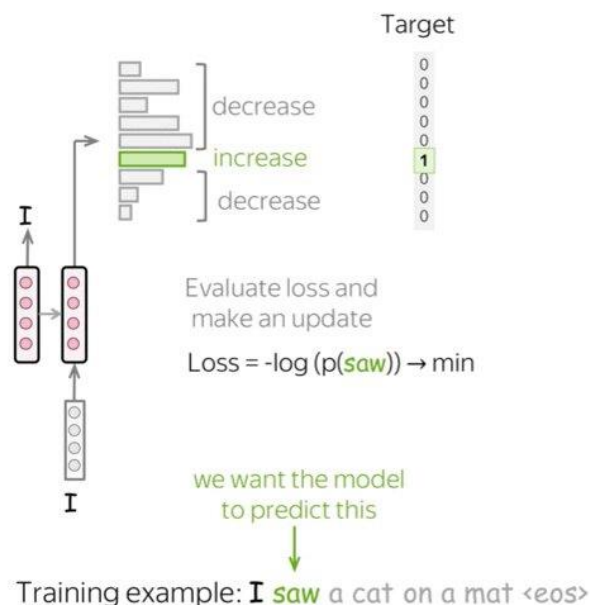
Language Model

- Model the generative likelihood of word sequences to predict the probabilities of future (or missing) tokens



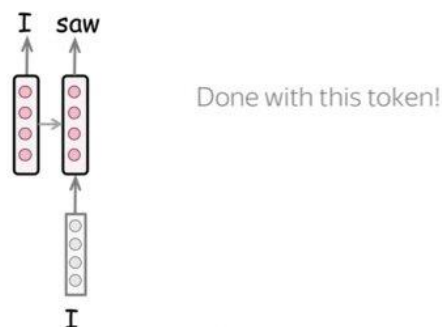
Language Model

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Language Model

- Model the generative likelihood of word sequences to predict the probabilities of future (or missing) tokens

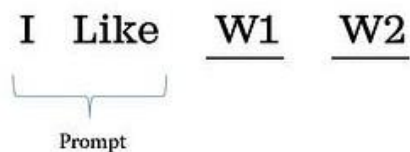


Training example: **I** **saw** a cat on a mat <eos>

Conditional text generation

- Example

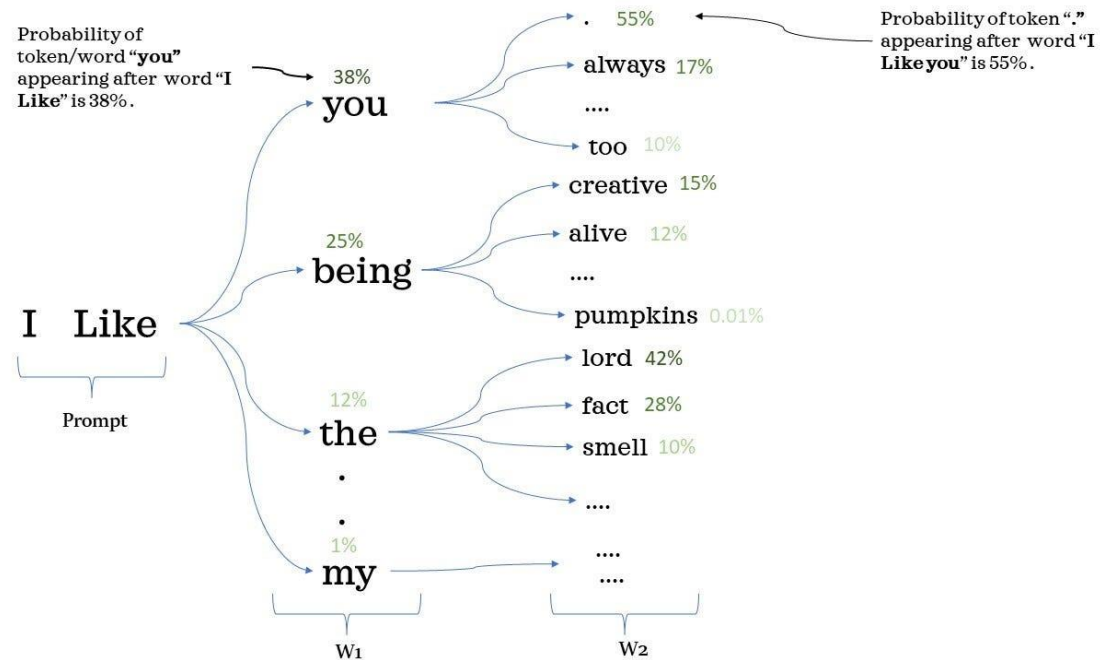
I Like W1 W2



The diagram illustrates a conditional text generation process. It shows a prompt "I Like" followed by two possible completions, "W1" and "W2". A bracket under "I Like" is labeled "Prompt".

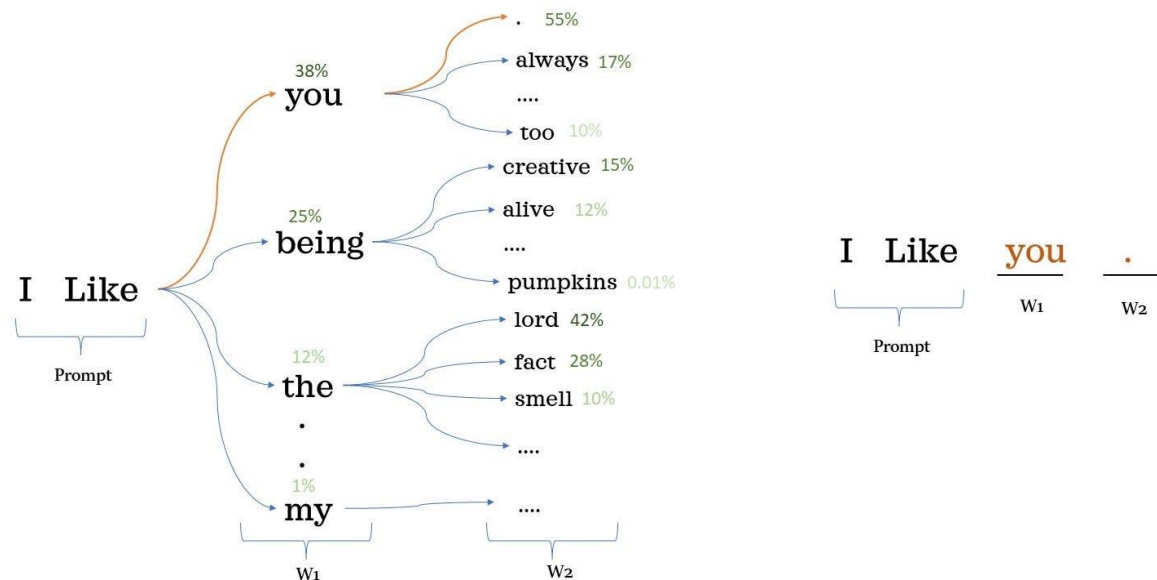
Conditional text generation

- Example



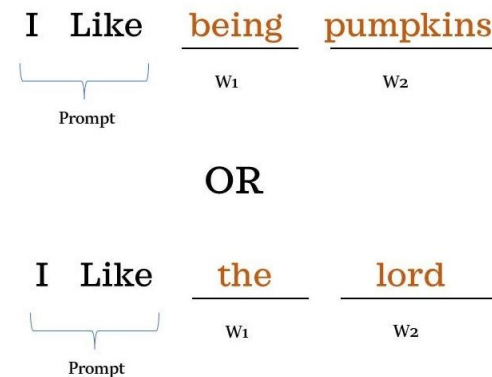
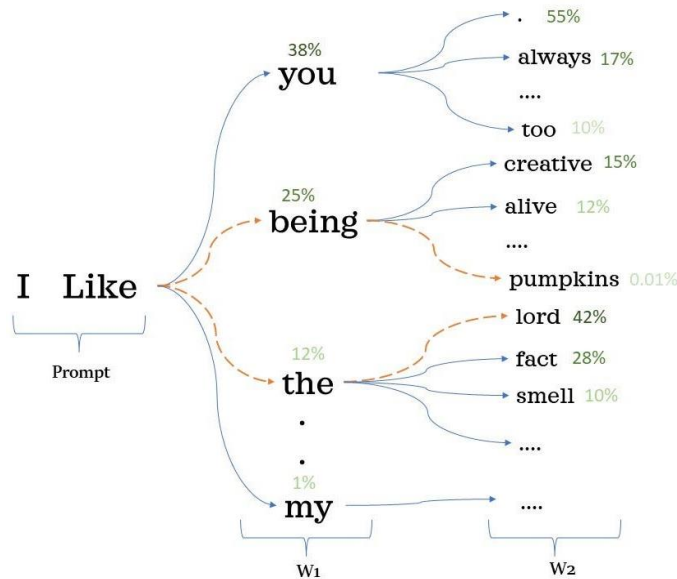
Conditional text generation

- $T = 0$



Conditional text generation

- $T > 0$



What is a Large Language Model?

- A groundbreaking NLP system with advanced language understanding and generation capabilities.
- Enables sophisticated text generation, question answering and conversational abilities.
- Trained on massive amounts of data to comprehend and perform a wide range of tasks.
- Revolutionizes information access, creativity, and natural language processing tasks.

What is a Large Language Model?

The content of the previous slide was generated by ChatGPT

What is a Large Language Model?

Provide me with a list of 4 short sentences (maximum 15 words each) to describe it



Free Research Preview. ChatGPT may produce inaccurate information about people, places, or facts. [ChatGPT May 24 Version](#)

Key LLM features

- Huge Deep Learning models
 - Billions of hyperparameters
- Virtually all state-of-the-art LLMs are based on Transformers
- Trained on massive data
 - TeraBytes of data
 - Trillions of tokens
 - Variety of sources, languages, domains, etc.
- Pretrained using self-supervised and/or semi-supervised learning
 - Self-supervised task: Next Token Prediction

Main LLM categories

- Proprietary vs. opensource
- Model
 - Characteristics
 - Complexity (number of hyperparameters)
- Training data
 - Language-specific vs. multilingual
 - Domain-specific vs. general-purpose

Main LLM categories

- Proprietary vs. opensource



Main LLM categories

- Language support
 - Apparently great but...

Model	Model Type			Model Sizes	Datasets	# Languages	LLM Objectives
	Enc-Only	Dec-Only	Enc-Dec				
XLM-R (Conneau et al., 2020)	✓			250M, 560M	CommonCraw	100	MLM
mBERT (Devlin et al., 2019)	✓			178M	Wikipedia	104	MLM
mDeBERTa v3 (He et al., 2021)	✓			278M	CC100	100	RTD w/ GDES
mBART-50 (Tang et al., 2020)			✓	611M	CC25, ML50	50	RTD w/ GDES
XGLM (Lin et al., 2021)		✓		564M, 1.7B, 2.9B, 4.5B, 7.5B	CommonCraw	30	MLM
BLOOMZ (Muennighoff et al., 2022)		✓		560M, 1.1B, 1.7B, 3B, 7.1B	ROOTS, xP3	46	Instruction Tuned
mT0 (Muennighoff et al., 2022)			✓	300M, 580M, 1.2B, 3.7B, 13B	mC4, xP3	~120	Instruction Tuned
ChatGPT (Bang et al., 2023)		✓		-	-	-	RLHF

Main LLM categories

- Language support
 - What's behind this?

Language	Percent	Language	Percent
en	89.70%	uk	0.07%
unknown	8.38%	ko	0.06%
de	0.17%	ca	0.04%
fr	0.16%	sr	0.04%
sv	0.15%	id	0.03%
zh	0.13%	cs	0.03%
es	0.13%	fi	0.03%
ru	0.13%	hu	0.03%
nl	0.12%	no	0.03%
it	0.11%	ro	0.03%
ja	0.10%	bg	0.02%
pl	0.09%	da	0.02%
pt	0.09%	sl	0.01%
vi	0.08%	hr	0.01%

Which languages does ChatGPT (proprietary LLM) understand?

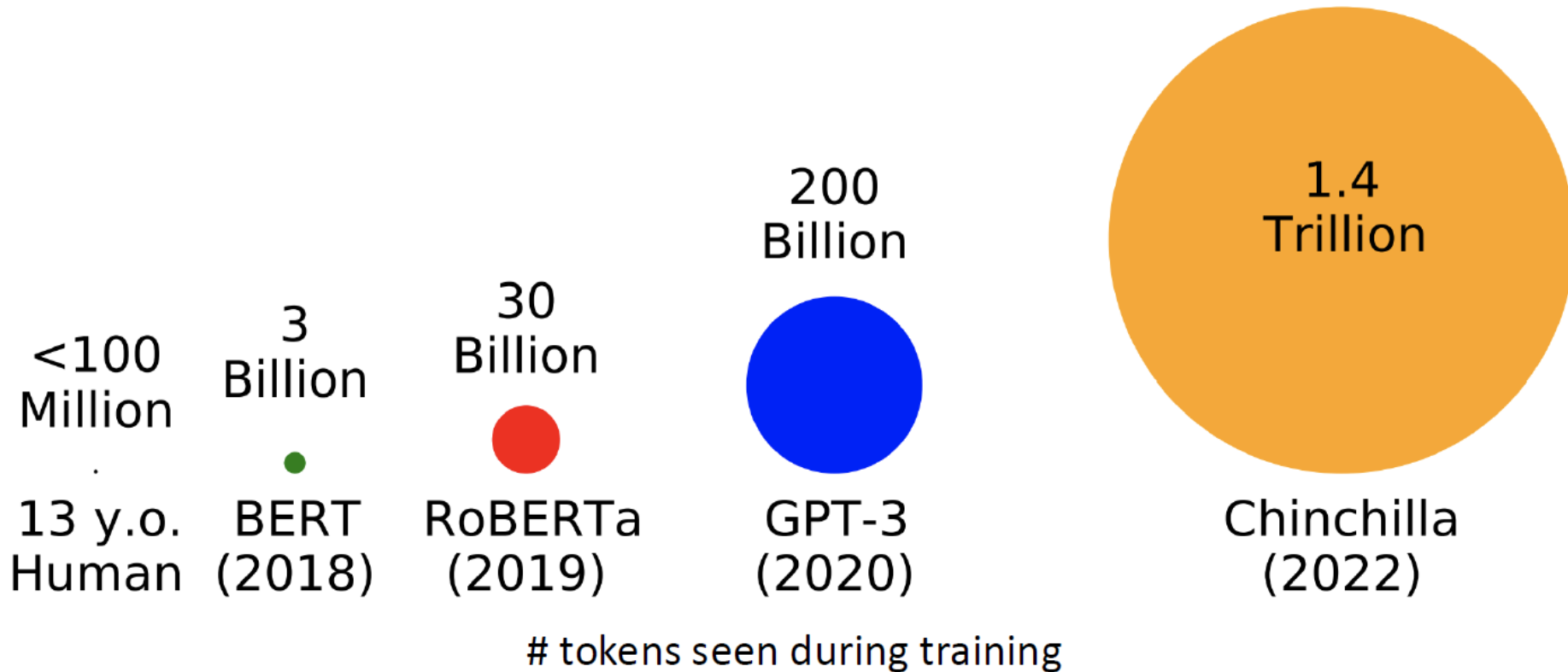
Albanian, Albania
Arabic, Arab World
Armenian, Armenia
Awadhi, India
Azerbaijani, Azerbaijan
Bashkir, Russia
Basque, Spain
Belarusian, Belarus
Bengali, Bangladesh
Bhojpuri, India
Bosnian, Bosnia and Herzegovina
Brazilian Portuguese, Brazil
Bulgarian, Bulgaria
Cantonese (Yue), China
Catalan, Spain
Chhattisgarhi, India
Chinese, China
Croatian, Croatia
Czech, Czech Republic
Danish, Denmark
Dogri, India
Dutch, Netherlands
English, United Kingdom
Estonian, Estonia
Faroese, Faroe Islands
Finnish, Finland
French, France
Galician, Spain
Georgian, Georgia
German, Germany

Greek, Greece
Gujarati, India
Haryanvi, India
Hindi, India
Hungarian, Hungary
Indonesian, Indonesia
Irish, Ireland
Italian, Italy
Japanese, Japan
Javanese, Indonesia
Kannada, India
Kashmiri, India
Kazakh, Kazakhstan
Konkani, India
Korean, South Korea
Kyrgyz, Kyrgyzstan
Latvian, Latvia
Lithuanian, Lithuania
Macedonian, North Macedonia
Maithili, India
Malay, Malaysia
Maltese, Malta
Mandarin, China
Mandarin Chinese, China
Marathi, India
Marwari, India
Min Nan, China
Moldovan, Moldova
Mongolian, Mongolia

Montenegrin, Montenegro
Nepali, Nepal
Norwegian, Norway
Oriya, India
Pashto, Afghanistan
Persian (Farsi), Iran
Polish, Poland
Portuguese, Portugal
Punjabi, India
Rajasthani, India
Romanian, Romania
Russian, Russia
Sanskrit, India
Santali, India
Serbian, Serbia
Sindhi, Pakistan
Sinhala, Sri Lanka
Slovak, Slovakia
Slovene, Slovenia
Slovenian, Slovenia
Ukrainian, Ukraine
Urdu, Pakistan
Uzbek, Uzbekistan
Vietnamese, Vietnam
Welsh, Wales
Wu, China

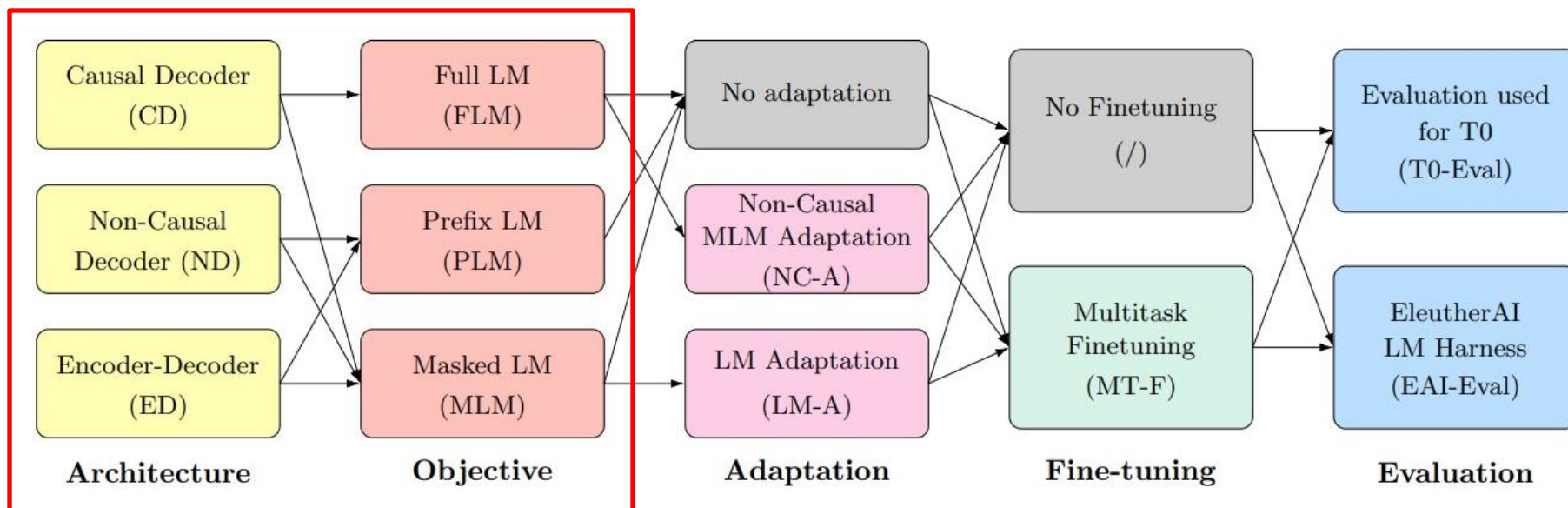
<https://botpress.com/blog/list-of-languages-supported-by-chatgpt>

LLMs complexity



<https://babylm.github.io/>

LLM architectures



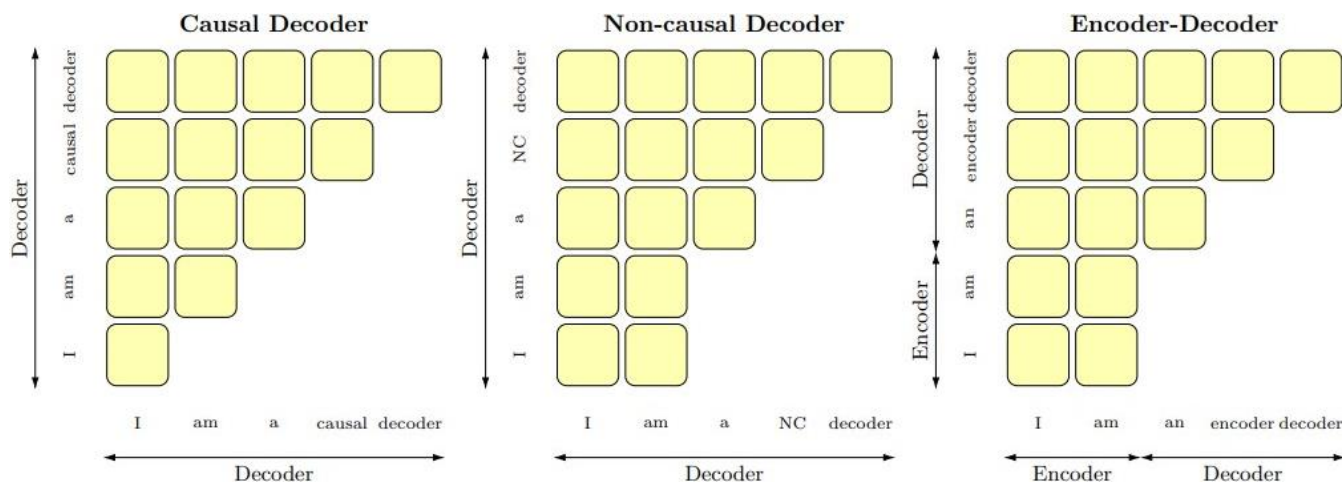
What Language Model Architecture and Pretraining Objective Work Best for Zero-Shot Generalization? <https://arxiv.org/abs/2204.05832>

LLM architectures

- Encoder-decoder vs. decoder-only
- All state-of-the-art language models over 100 billions of parameters are **autoregressive decoder-only models**
 - They achieve better **zero-shot** performance than encoder-decoder architectures

LLM architectures

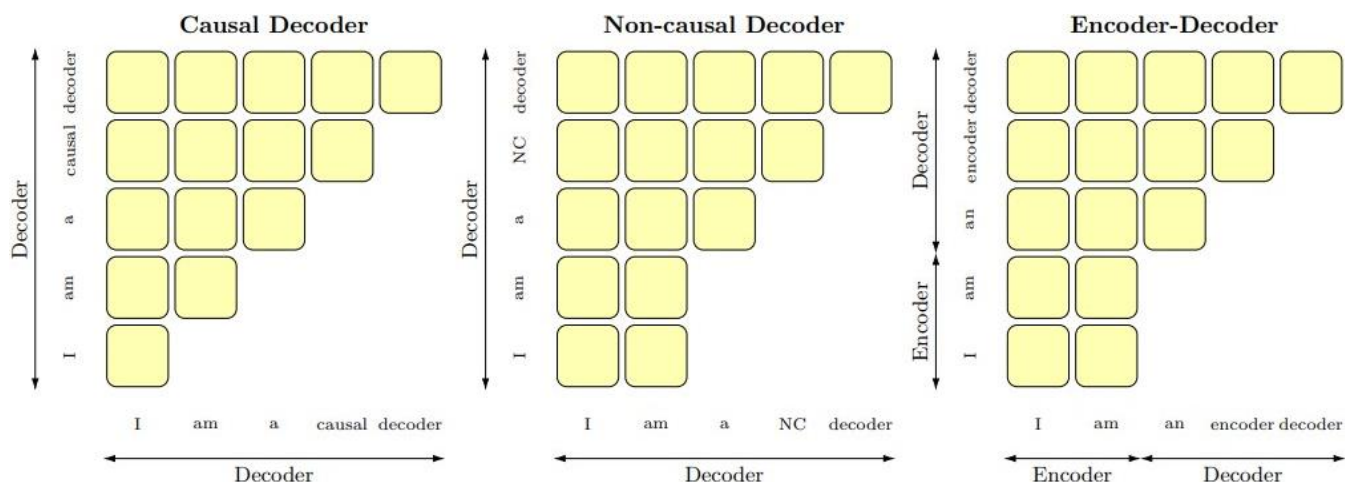
- **Causal Decoder-only:** conditioning is based on past tokens only
 - Predict the next token in the sequence
 - All tokens are processed in an equivalent fashion
 - Backbone of GPT series



What Language Model Architecture and Pretraining Objective Work Best for Zero-Shot Generalization? <https://arxiv.org/abs/2204.05832>

LLM architectures

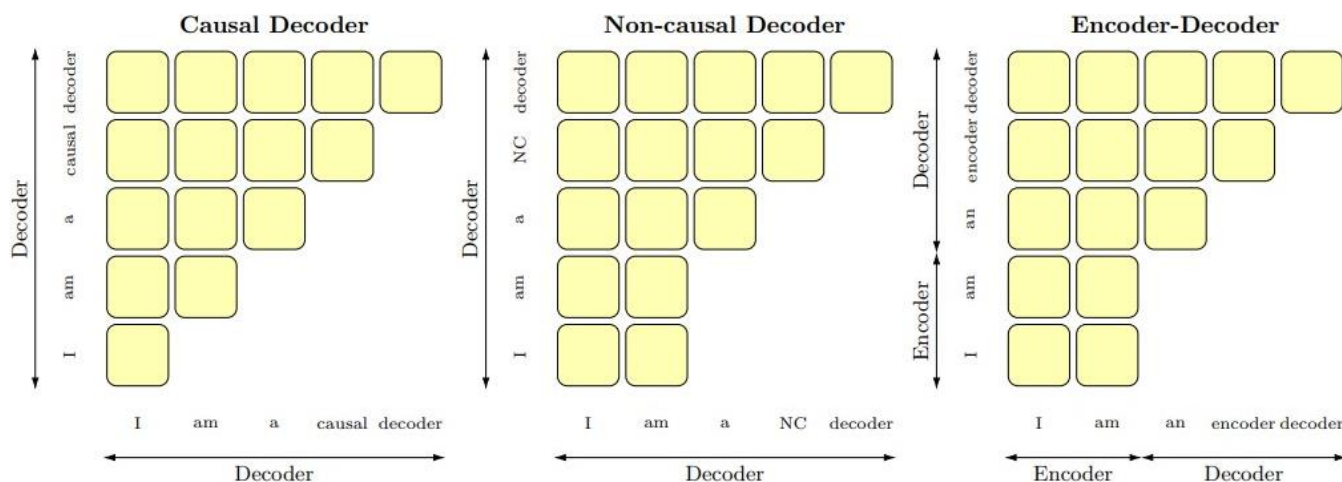
- **Non-causal Decoder-only:** the self-attention mask is modified so that attention is not restricted to past tokens
 - The region of the input sequence corresponding to conditioning information has a non-causal mask (eventually not restricted to past tokens)
 - Limited adoption



What Language Model Architecture and Pretraining Objective Work Best for Zero-Shot Generalization? <https://arxiv.org/abs/2204.05832>

LLM architectures

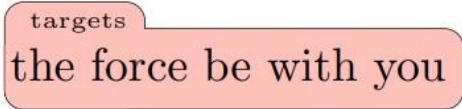
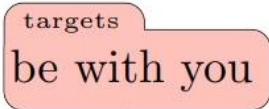
- **Encoder-Decoder:** the prediction of the target sequence is conditioned on the output of the encoder
 - relies on cross-attention (such as BART or T5)
 - Based on BERT-like encoders



What Language Model Architecture and Pretraining Objective Work Best for Zero-Shot Generalization? <https://arxiv.org/abs/2204.05832>

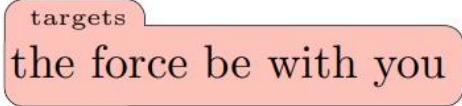
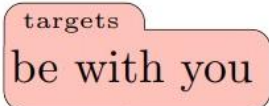
LLM pre-training objectives

- **Full Language Modeling:** given the previous tokens, the model is trained to predict the following one

Full Language Modeling	May 
Prefix Language Modeling	May the force 
Masked Language Modeling	May  be with you

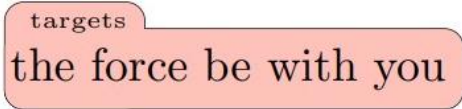
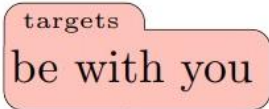
LLM pre-training objectives

- **Prefix Language Modeling:** define a prefix where the attention mask is allowed to be non-causal
 - The model is trained to predict each token outside the prefix given all previous tokens
 - For inference, the prefix is the input text
 - For pretraining, the prefix is commonly chosen randomly

Full Language Modeling	May 
Prefix Language Modeling	May the force 
Masked Language Modeling	May  be with you

Pre-training objectives

- **Masked Language Modeling:** (spans of) tokens in the input text are replaced/corrupted with a special mask token
 - The model is trained to predict/reconstruct the missing/ corrupted tokens
 - Text span masking or corruption as in BART
 - For pretraining, the prefix is commonly chosen randomly

Full Language Modeling	May 
Prefix Language Modeling	May the force 
Masked Language Modeling	May  be with you

LLM adaptation

- The process of taking a LM and turning it into a task model
- It takes as input
 - A natural language description
 - A set of training instances (input-output pairs)



<https://docs.sia.so/sia/llmops/what-steps-are-involved-in-llmops/adaptation-to-downstream-tasks>

LLM adaptation: training

- **Training via supervised learning:** train a new model that maps inputs to outputs, either by
 - **Probing:** creating a new task-specific model that uses the language model as latent features
 - Train on top of a frozen LM
 - **Fine-tuning:** starting with the language model and updating it based on the training instances
 - **Lightweight fine-tuning:** optimize just few parameters
 - A mix of fine-tuning and probing
- Challenged by data overfitting
 - Few training instances compared to the network complexity

LLM adaptation: training

- Rationale

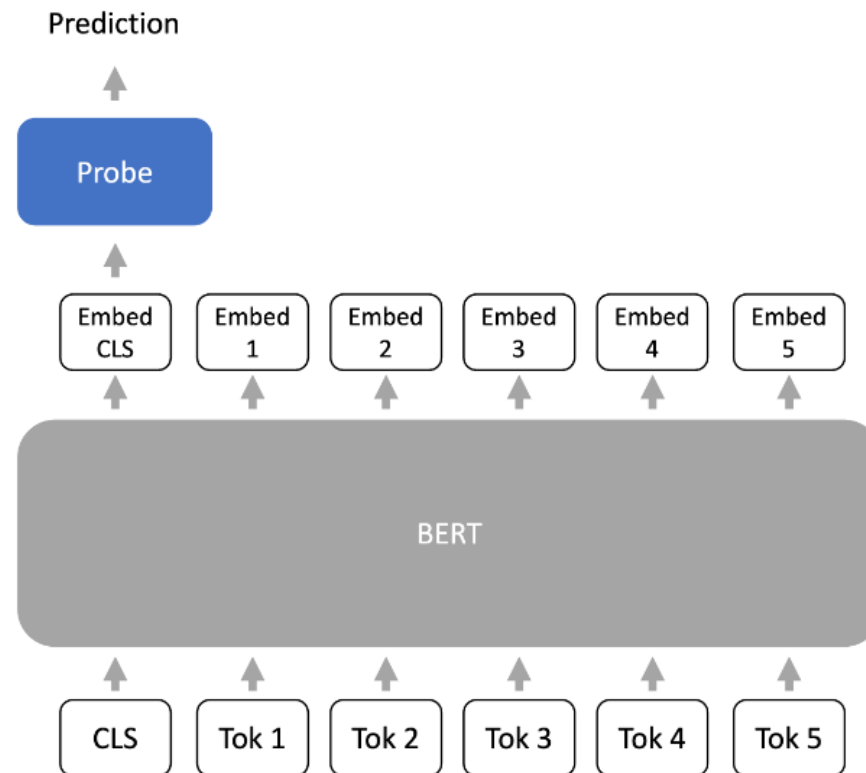
- LLMs are trained in a task-agnostic way
- Downstream tasks can be various: next token prediction can be not sufficient!

- Main strategies

- **Formatting:** use a particular formatting style tailored to the downstream task
- **Topic shift:** adapt the model to a specific domain
- **Temporal shift:** when the downstream task requires new model that was not available during pre-training

LLM adaptation: training

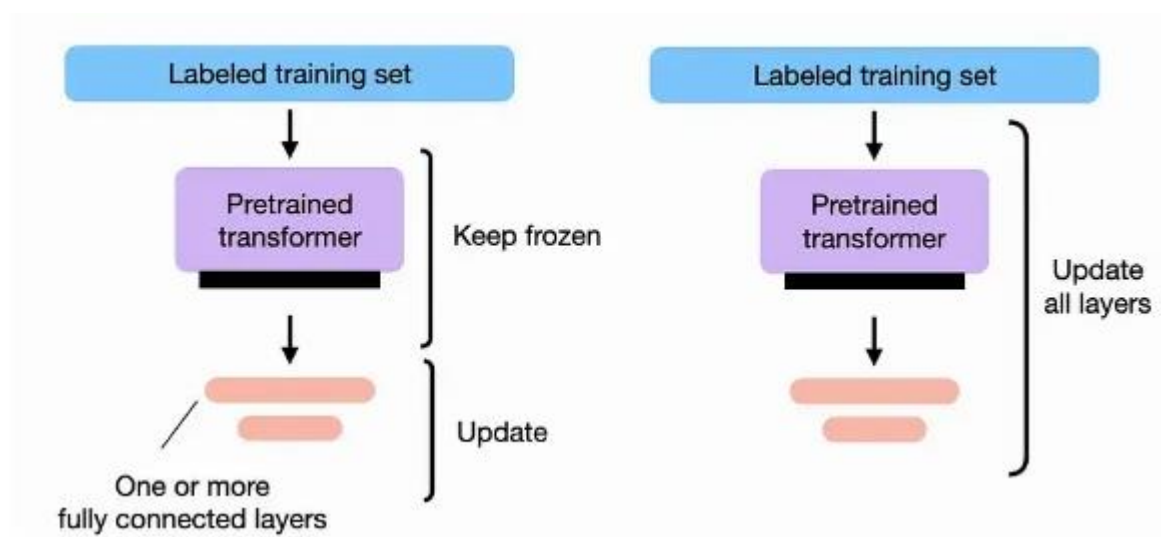
- Example of **probing**: train a prediction head from the last layer of representation of the LM to the output



<https://stanford-cs324.github.io/winter2022/lectures/capabilities/>

LLM adaptation: training

- Examples of **fine-tuning** and **lightweight fine-tuning** via supervised learning

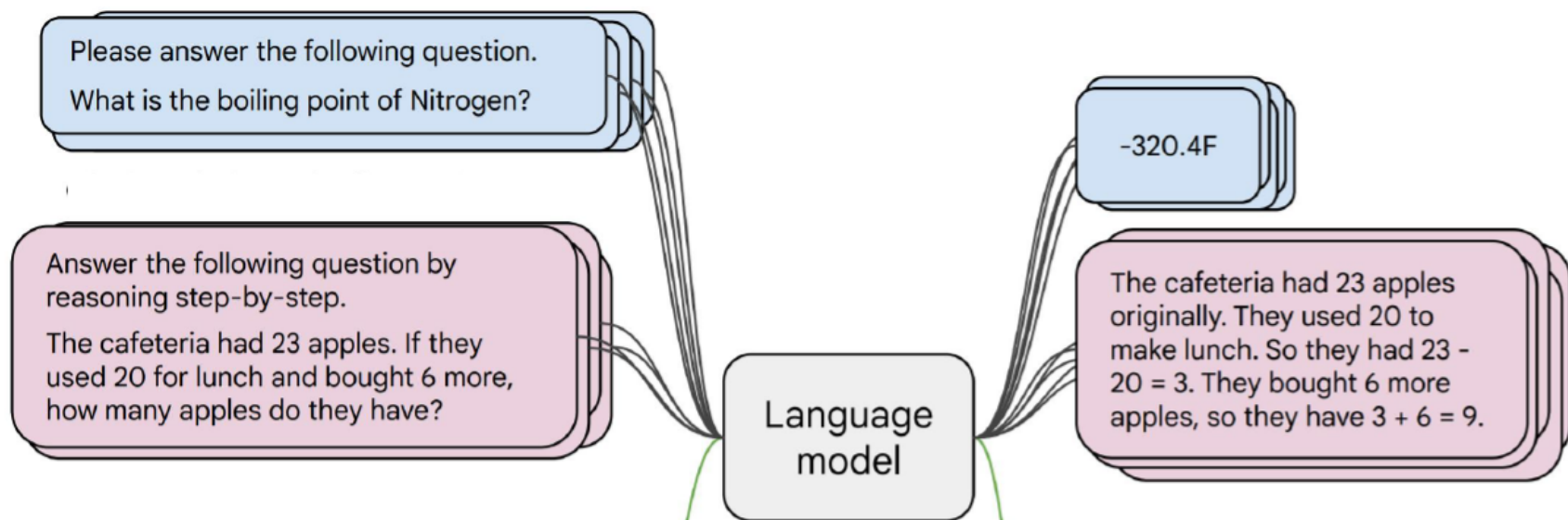


LLM adaptation: training

- **Fine-tuning:** uses the pretrained language model parameters as initialization for optimization
 - Requires storing a LLM specialized for every downstream task
 - Expensive but more effective than probing
 - Can rely on **Reinforcement Learning** as well to account human preferences
- **Lightweight fine-tuning:** achieve the same expressivity as full fine-tuning while not requiring us to store the full language model for every task

Instruction fine-tuning

Collect examples of (instruction, output) pairs across many tasks and finetune an LM



Evaluate on **unseen tasks**

Q: Can Geoffrey Hinton have a conversation with George Washington?
Give the rationale before answering.

Geoffrey Hinton is a British-Canadian computer scientist born in 1947. George Washington died in 1799. Thus, they could not have had a conversation together. So the answer is "no".

Flan-T5 Chung et al., 2022

Which tasks?



Wang et al., 2022

Drawbacks of instruction fine-tuning

- Cost of collecting ground-truth data
- Open-ended creative generation is not feasible
- Language modeling penalizes all token-level mistakes equally, but some errors are worse than others

Reinforcement Learning from Human Feedback

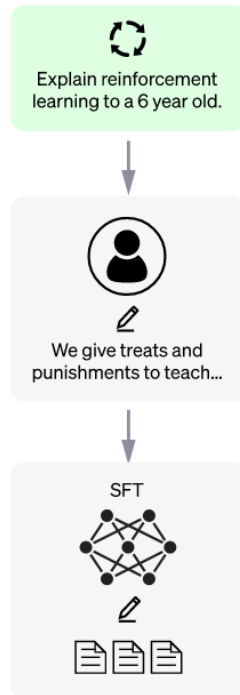
Step 1

Collect demonstration data and train a supervised policy.

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3.5 with supervised learning.



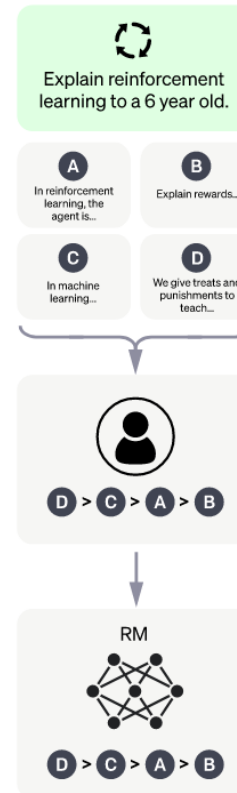
Step 2

Collect comparison data and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

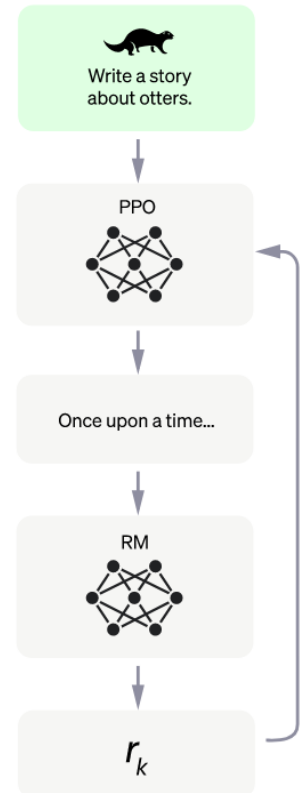
A new prompt is sampled from the dataset.

The PPO model is initialized from the supervised policy.

The policy generates an output.

The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.



Source: <https://openai.com/blog/chatgpt>

RLHF Step 1: Supervised Fine-Tuning

- Human trainers provide conversations in which they play both sides (user and AI assistant)
- Give the trainers access to model-written suggestions to help them compose their responses
- New human-collected datasets

RLHF Step 2: Reinforcement Learning

- Collect comparison data by randomly selecting a model-written message and sampling several alternative completions (from previously collected conversations)
- Human trainers rank responses
- Create “reward models” from human rankings

Reward model

- Initialized with a pretrained model
- A regressor returning the score of any given prompt and response
- Based on cross-entropy minimization:

$$\mathcal{L}(\theta) = -\frac{1}{\binom{K}{2}} E_{(x, y_w, y_l)} [\log(\sigma(r_\theta(x, y_w) - r_\theta(x, y_l)))] = -\frac{1}{\binom{K}{2}} E_{(x, y_w, y_l)} \log \left[\frac{e^{r_\theta(x, y_w)}}{e^{r_\theta(x, y_w)} + e^{r_\theta(x, y_l)}} \right]$$

where

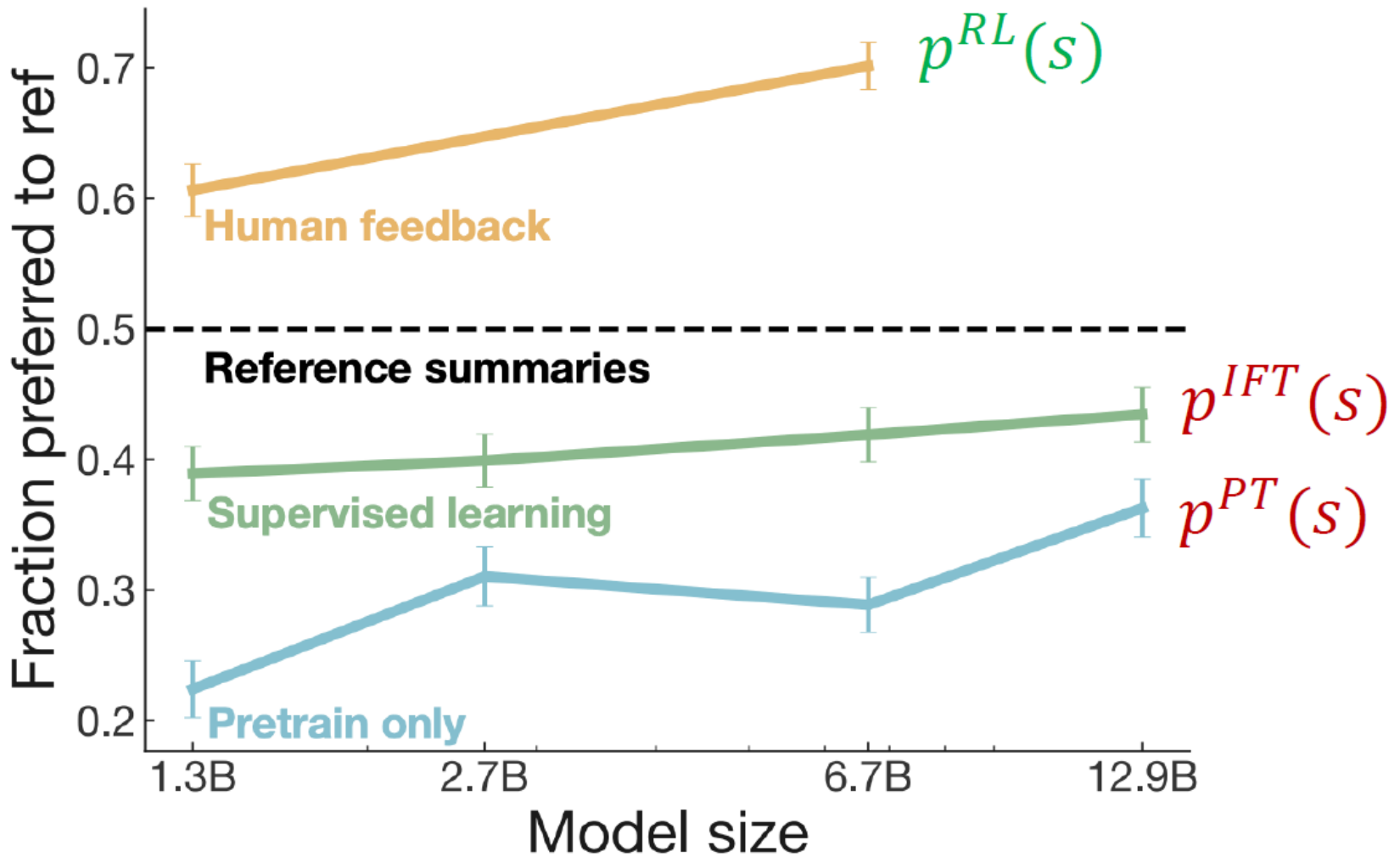
K: number of responses, x: prompt, y: response, E[X]: expected value

- It predicts the probability that response y_w is preferred to y_l

RLHF Step 3: Proximal Policy Optimization

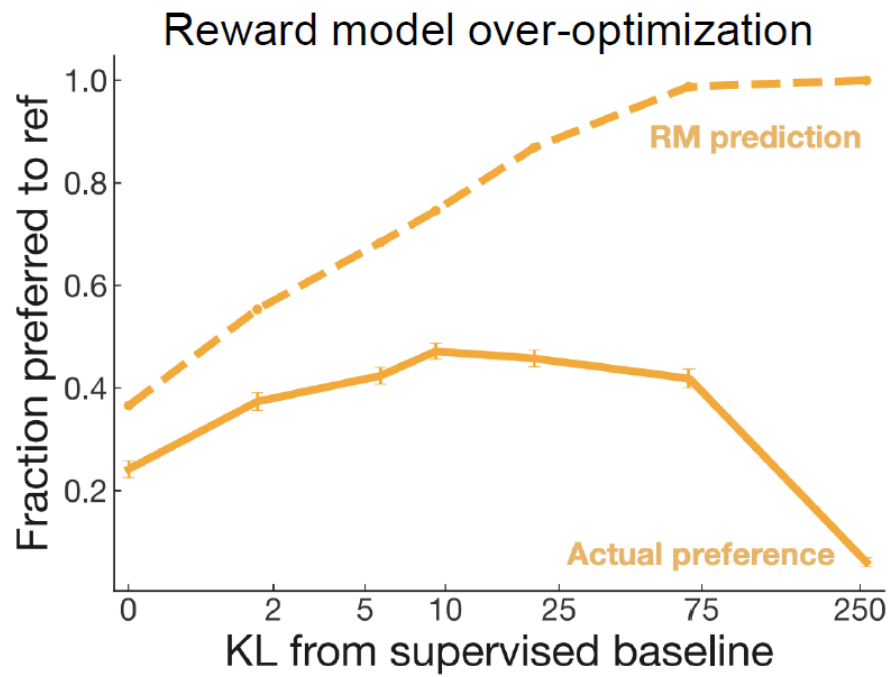
- The reward model is used to fine-tune the model further by several iterations of Proximal Policy Optimization (PPO)
 - The objective function calculates how well the policy's responses are expected to align with human feedback.
 - The policy generates responses to prompts, and each response is evaluated both on how well it matches human preferences (as measured by the reward model) and how similar it is to responses the model would naturally generate.
 - The goal is to balance improving alignment with human preferences while ensuring the model's responses remain diverse and not too far removed from what it has learned during its initial training.
 - This helps the model not only to provide answers that people find useful or agreeable but also to maintain a broad understanding and avoid overly narrow or repetitive responses.

RLHF contribution



RLHF drawbacks

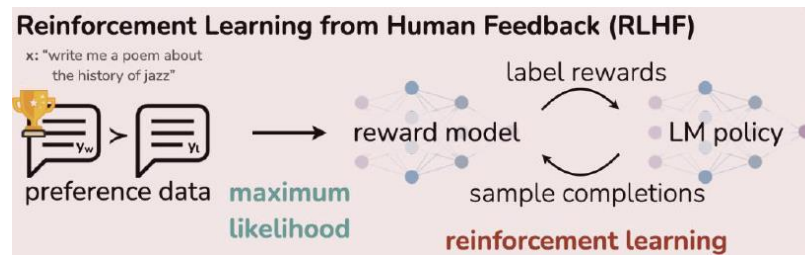
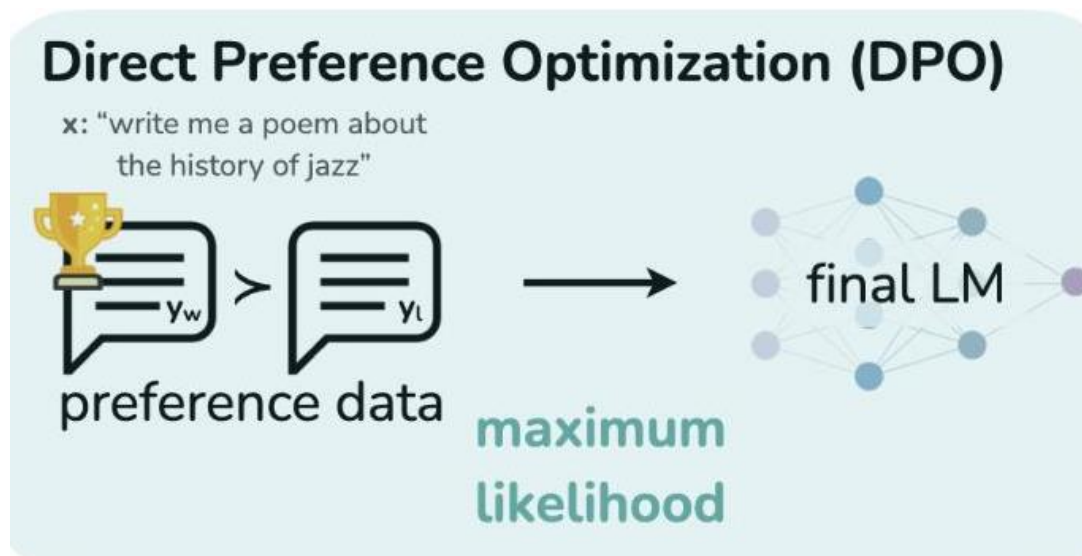
- Human preferences are unreliable
 - Reward hacking
 - Hallucination



Stiennon et al. 2020

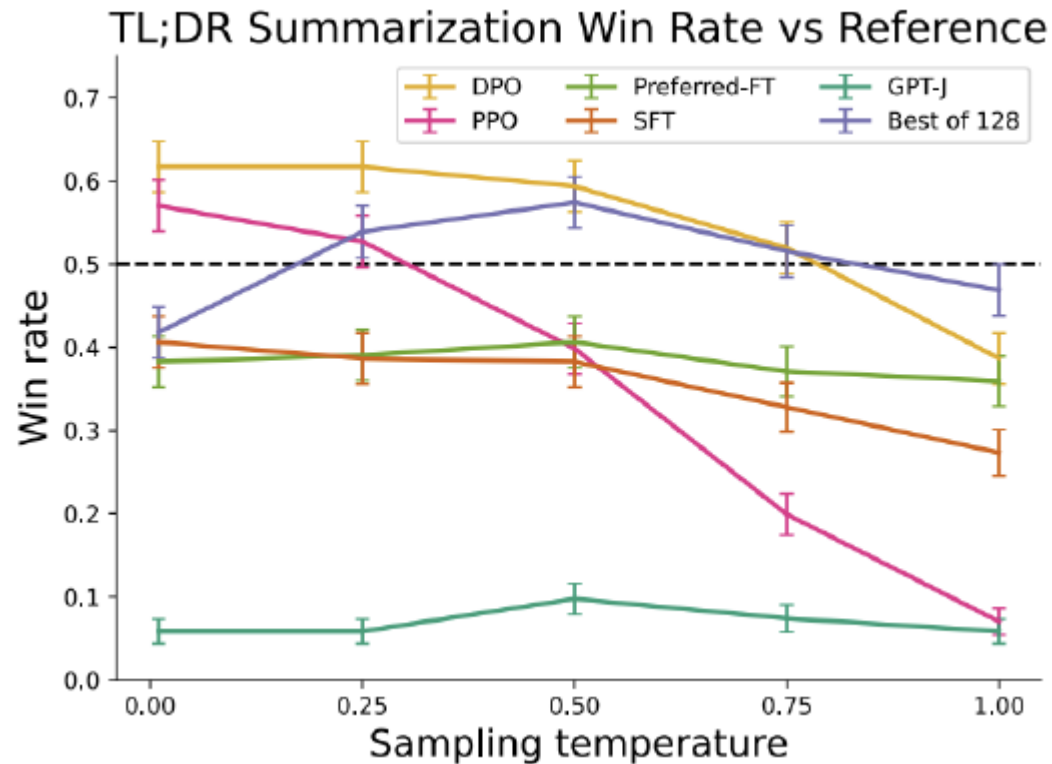
Direct Preference Optimization

- Replace RL with a simpler weighted Maximum Likelihood Estimation objective



Stiennon et al. 2020

DPO advantage



Rafailon et al. 2023

RLHF: formal statement

1. **Supervised Fine-Tuning:** produces a model

$$\pi^{\text{SFT}}$$

2. **Reward Model Fitting:** Given the pairs

$$(y_1, y_2) \sim \pi^{\text{SFT}}(y \mid x)$$

ask human labelers preferences about preferred and dispreferred responses

$$y_w \succ y_l \mid x$$

And fit a reward model

$$r^*(y, x)$$

<https://arxiv.org/pdf/2305.18290>

RLHF

Distribution of human preferences (Bradley-Terry model)

$$p^*(y_1 \succ y_2 \mid x) = \frac{\exp(r^*(x, y_1))}{\exp(r^*(x, y_1)) + \exp(r^*(x, y_2))}$$

Given a dataset of comparisons sampled from the distribution p^*

$$\mathcal{D} = \{x^{(i)}, y_w^{(i)}, y_l^{(i)}\}_{i=1}^N$$

The reward model $r_\phi(x, y)$ can be parametrized and estimated via Maximum Likelihood with loss

$$\mathcal{L}_R(r_\phi, \mathcal{D}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} [\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))]$$

<https://arxiv.org/pdf/2305.18290>

RLHF

Initialization phase:

$$r_{\phi}(x, y) = \pi^{\text{SFT}}(y \mid x)$$

with the addition of a linear layer on top of the final transformer layer that produces a single scalar prediction for the reward value

Rewards are normalized:

$$\mathbb{E}_{x, y \sim \mathcal{D}} [r_{\phi}(x, y)] = 0 \text{ for all } x.$$

RLHF

3) RL policy optimization

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(y|x)} [r_{\phi}(x, y)] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(y | x) || \pi_{\text{ref}}(y | x)]$$

Where

$$\pi_{\theta} = \pi^{\text{SFT}}$$

and

β is the parameter controlling the deviation from the initial policy.

RLHF

Due to the discrete nature of language, the reward is defined by

$$r(x, y) = r_{\phi}(x, y) - \beta(\log \pi_{\theta}(y \mid x) - \log \pi_{\text{ref}}(y \mid x))$$

to allow PPO optimization.

DPO

- **Key idea**

- the policy network represents both the language model and the (implicit) reward
 - From a loss function over reward scores to loss function using policies
- Avoiding fitting a standalone reward model while still optimizing under existing models of human preferences

$$p^*(y_1 \succ y_2 \mid x) = \frac{1}{1 + \exp \left(\beta \log \frac{\pi^*(y_2|x)}{\pi_{\text{ref}}(y_2|x)} - \beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)} \right)}$$

where π^* is the optimal policy and π_{ref} is the reference policy.

DPO

- Policy objective

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$

- Loss gradient

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = \\ -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_l) - \hat{r}_{\theta}(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_{\theta} \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_{\theta} \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right] \end{aligned}$$

<https://arxiv.org/pdf/2305.18290>

DPO

- Implicit reward over the language and reference models

$$\hat{r}_\theta(x, y) = \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$$

Whenever available

$$\pi_{\text{ref}} = \pi^{\text{SFT}}$$

Otherwise

$$\pi_{\text{ref}} = \arg \max_{\pi} \mathbb{E}_{x, y_w \sim \mathcal{D}} [\log \pi(y_w | x)].$$

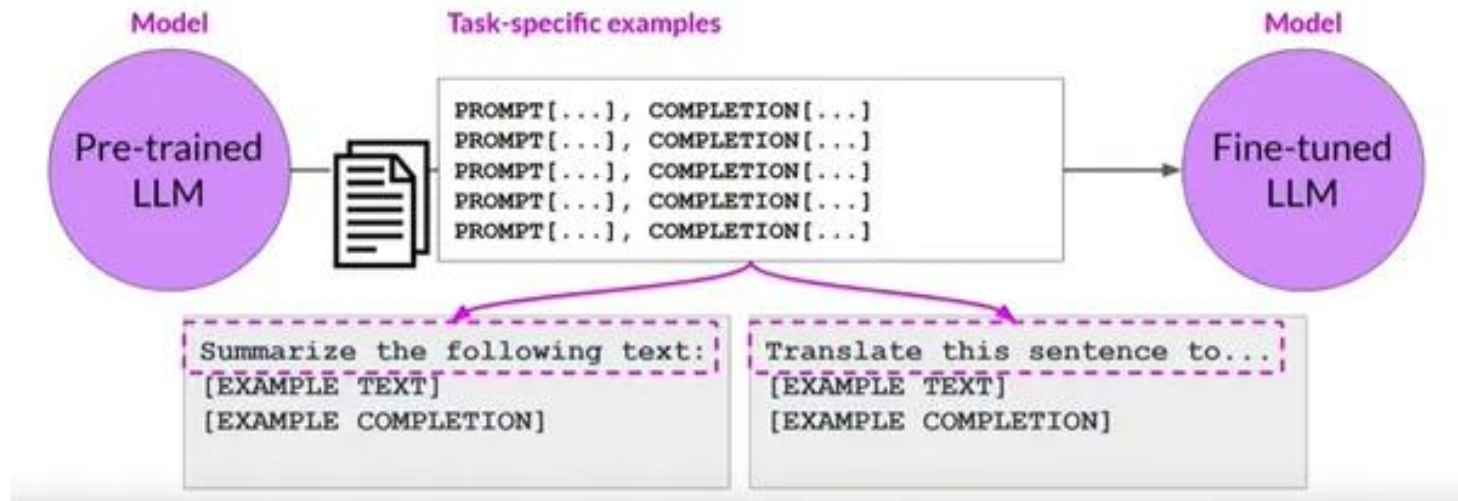
<https://arxiv.org/pdf/2305.18290>

LLM adaptation: prompting

- **Prompting (in-context learning):** Construct a prompt (a string based on the description and training instances) or a set of prompts, feed those into a language model to obtain completions.
 - **Zero-shot learning:** number of training examples is 0
 - **One-shot learning:** number of training examples is 1
 - **Few-shot learning:** number of training examples is few

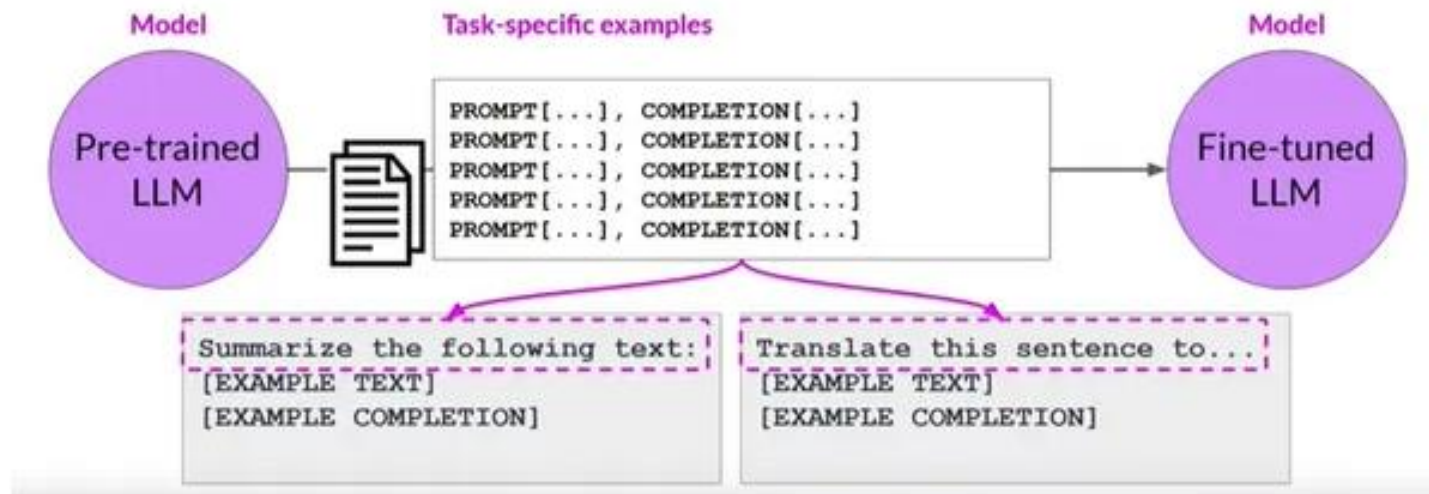
LLM adaptation: prompting

- **Prompting (in-context learning):** example of **single-task** for text summarization



LLM adaptation: prompting

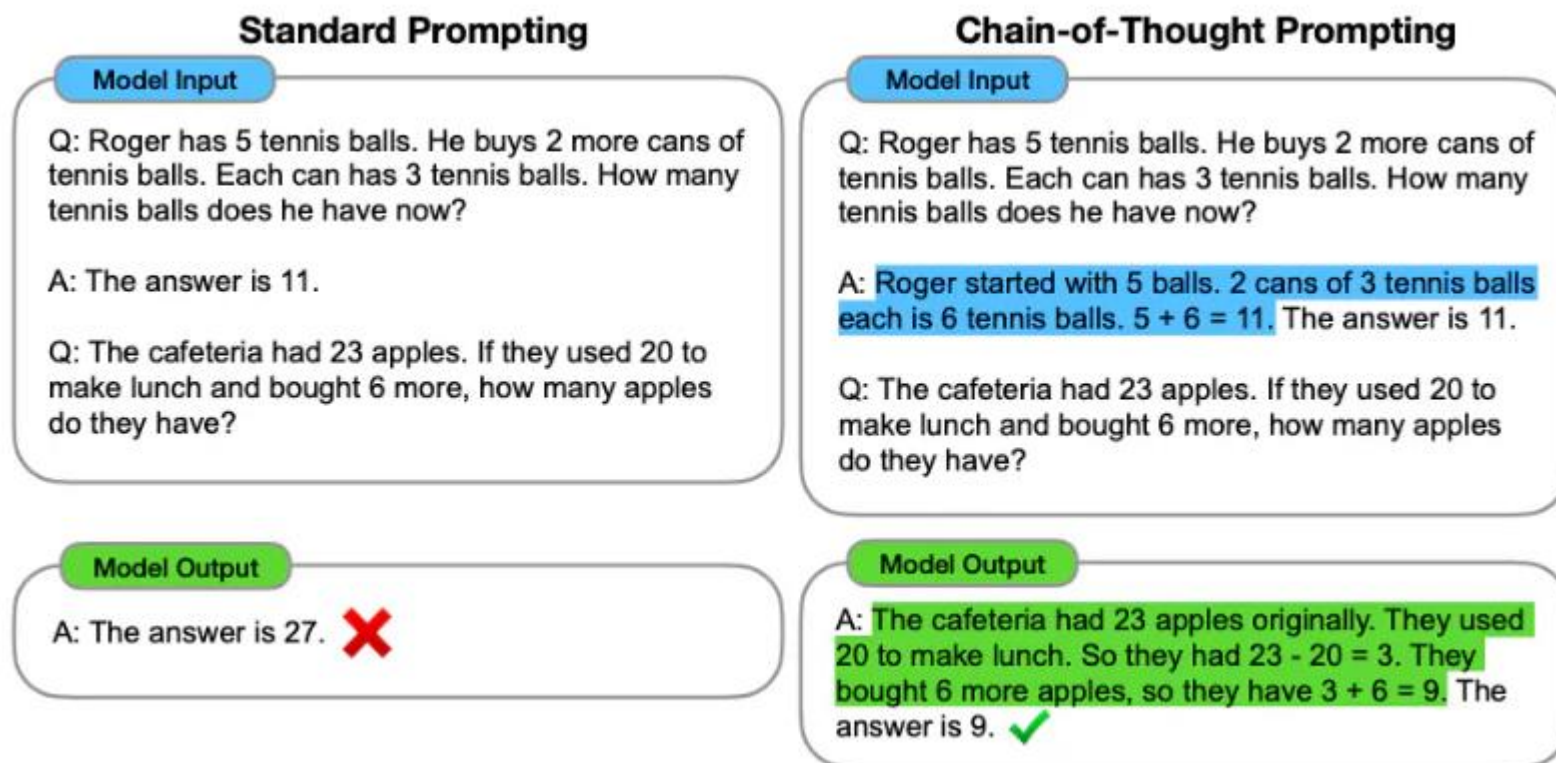
- **Prompting (in-context learning):** example of **multi-task** for text summarization and machine translation



<https://stanford-cs324.github.io/winter2022/lectures/capabilities/>

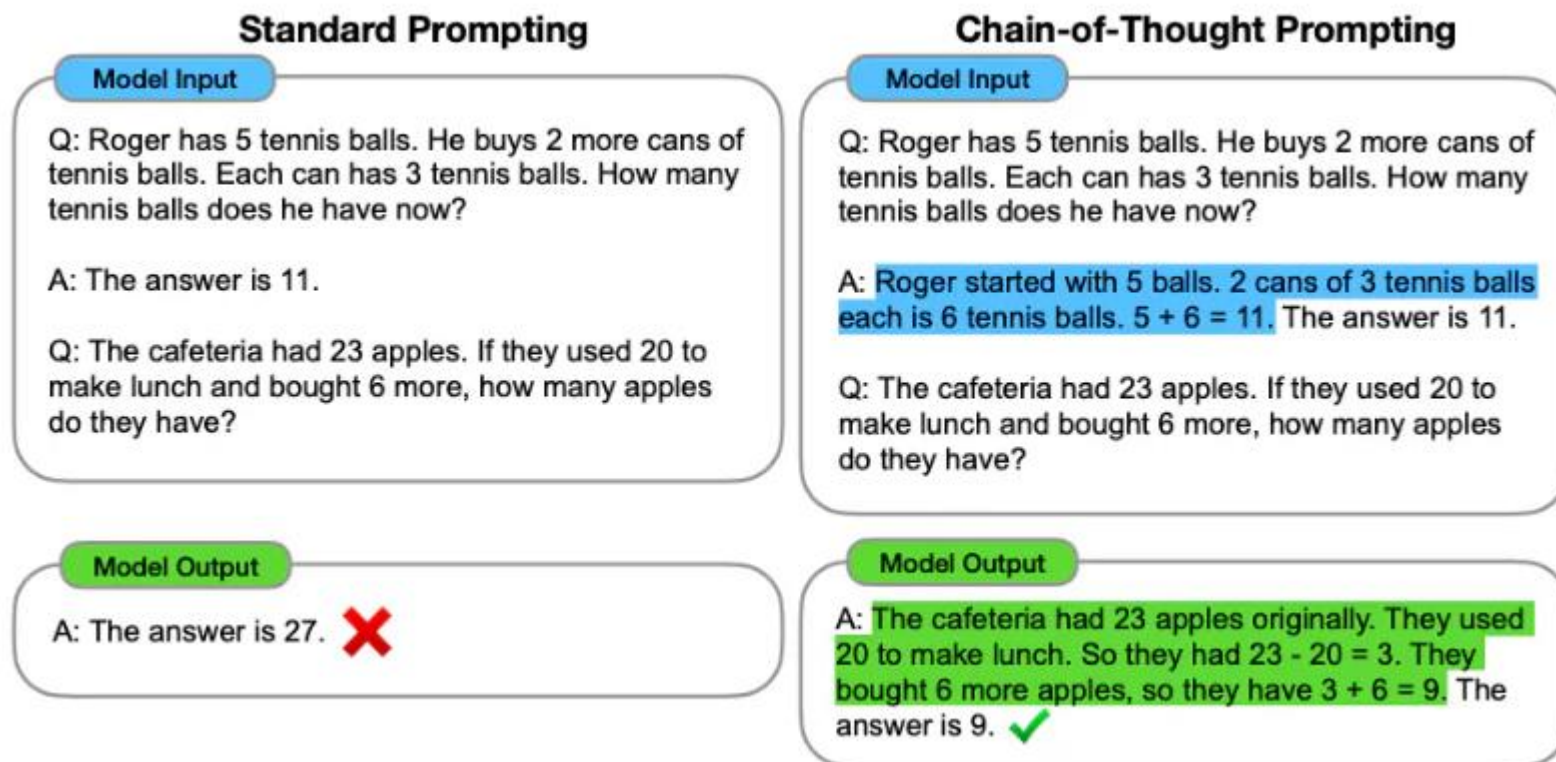
Chain-Of-Thoughts prompting

- Enable complex reasoning capabilities through intermediate reasoning steps.



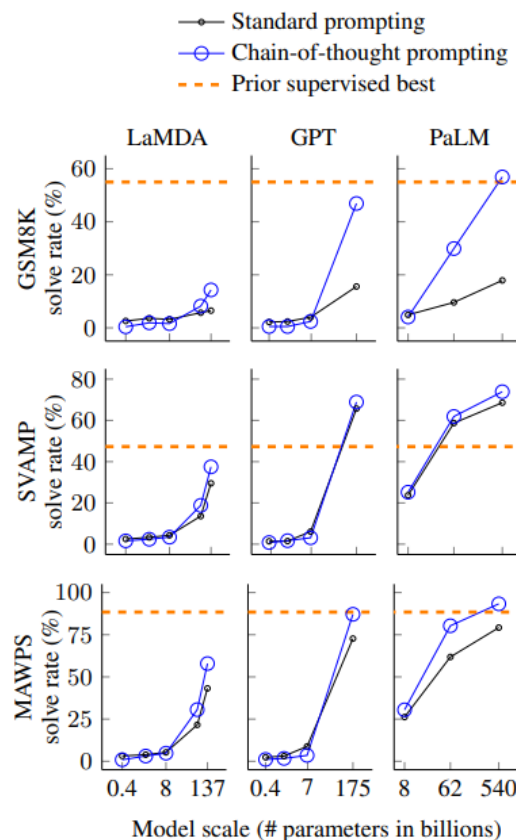
Chain-Of-Thoughts prompting

- Enable complex reasoning capabilities through intermediate reasoning steps
 - E.g., tackle complex arithmetic, commonsense, and symbolic reasoning tasks



Chain-Of-Thoughts prompting

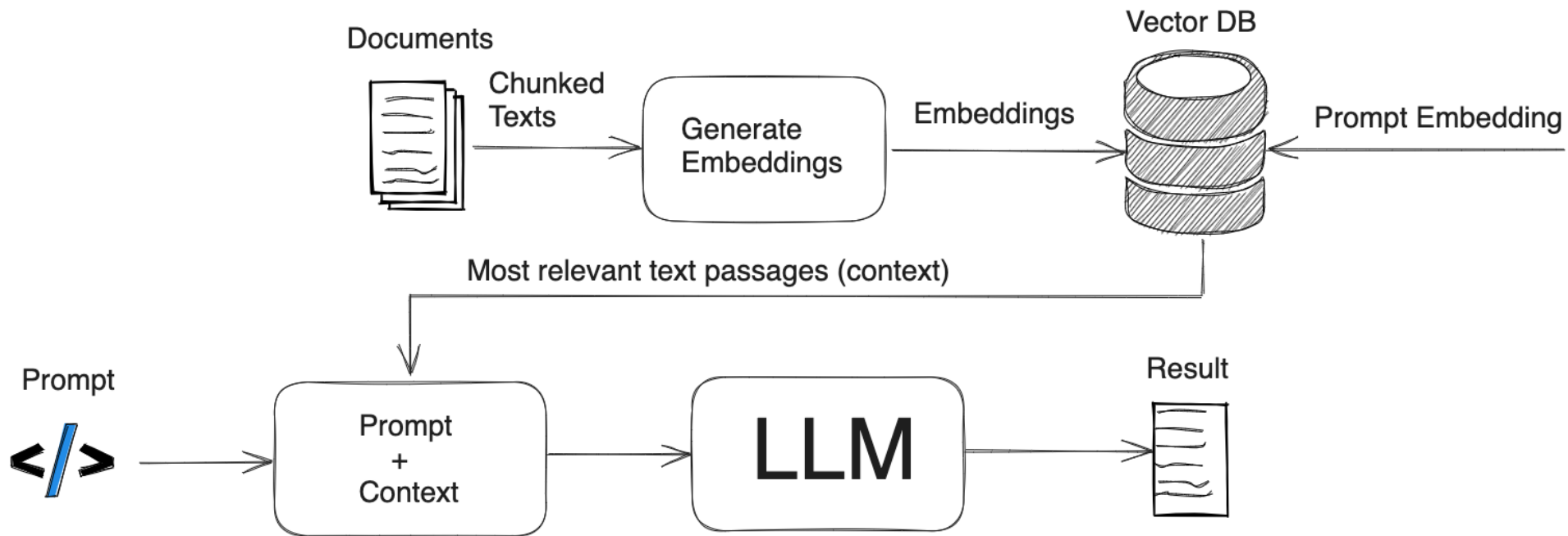
- Enable complex reasoning capabilities through intermediate reasoning steps
 - E.g., tackle complex arithmetic, commonsense, and symbolic reasoning tasks



Chain-Of-Thoughts prompting: examples of <input,COT,output>

Math Word Problems (free response) Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.	Math Word Problems (multiple choice) Q: How many keystrokes are needed to type the numbers from 1 to 500? Answer Choices: (a) 1156 (b) 1392 (c) 1480 (d) 1562 (e) 1788 A: There are 9 one-digit numbers from 1 to 9. There are 90 two-digit numbers from 10 to 99. There are 401 three-digit numbers from 100 to 500. $9 + 90(2) + 401(3) = 1392$. The answer is (b).	CSQA (commonsense) Q: Sammy wanted to go to where the people were. Where might he go? Options: (a) race track (b) populated areas (c) desert (d) apartment (e) roadblock A: The answer must be a place with a lot of people. Race tracks, desert, apartments, and roadblocks don't have a lot of people, but populated areas do. So the answer is (b).
StrategyQA Q: Yes or no: Would a pear sink in water? A: The density of a pear is about 0.6 g/cm^3, which is less than water. Thus, a pear would float. So the answer is no.	Date Understanding Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY? A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.	Sports Understanding Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship." A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.
SayCan (Instructing a robot) Human: How would you bring me something that isn't a fruit? Explanation: the user wants something to eat that isn't a fruit. An energy bar is not a fruit, so I will bring the user an energy bar. Plan: 1. find(energy bar) 2. pick(energy bar) 3. find(user) 4. put(energy bar) 5. done().	Last Letter Concatenation Q: Take the last letters of the words in "Lady Gaga" and concatenate them. A: The last letter of "Lady" is "y". The last letter of "Gaga" is "a". Concatenating them is "ya". So the answer is ya.	Coin Flip (state tracking) Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up? A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

Retrieval Augmented Generation



<https://ai.meta.com/blog/retrieval-augmented-generation-streamlining-the-creation-of-intelligent-natural-language-processing-models/>

Retrieval Augmented Generation

- Combine LLMs with content retrieval systems
- Augment the capabilities of LLMs using a hybrid approach
- Particularly relevant to companies that do not want to disclose enterprise data
- RAG looks and acts like a standard seq2seq model
 - They take in one sequence and outputs a corresponding sequence.
- There is an intermediary step though, which differentiates and elevates RAG above the usual seq2seq methods.
 - Rather than passing the input directly to the generator, RAG instead uses the input to retrieve a set of relevant documents

<https://ai.meta.com/blog/retrieval-augmented-generation-streamlining-the-creation-of-intelligent-natural-language-processing-models/>

Retrieval Augmented Generation

- Given the prompt “When did the first mammal appear on Earth?” for instance, RAG might surface documents for “Mammal,” “History of Earth,” and “Evolution of Mammals.” These supporting documents are then concatenated as context with the original input and fed to the seq2seq model that produces the actual output.
- RAG thus has two sources of knowledge: the knowledge that seq2seq models store in their parameters (parametric memory) and the knowledge stored in the corpus from which RAG retrieves passages (nonparametric memory).

Examples of famous LLMs

- GPT-* family (GPT-3, GPT-3.5, ChatGPT, GPT-4)
- LaMDA & LaMDA 2
- PaLM & PaLM 2
- BLOOM
- Galactica
- LLaMA
- ...

GPT-* family

- **G**enerative **P**re-trained **T**ransformers
- Developed by: OpenAI
- Release date:
 - GPT-3: May 2020 – GPT-3.5: March 2022
ChatGPT: November 2022 – GPT-4: March 2023
- Number of parameters:
 - GPT-3, GPT-3.5, ChatGPT: 175 billion
 - GPT-4: undisclosed, 1 trillion (estimated)

GPT-* family

- Training corpus:
 - GPT-3: 499 billion tokens
 - GPT-3.5, ChatGPT, GPT-4, GPT-5: undisclosed, trillion tokens (estimated)
- Paper:
 - GPT-3: <https://arxiv.org/abs/2005.14165>
 - GPT-3.5, ChatGPT: not available
 - GPT-4: <https://arxiv.org/abs/2303.08774>

Note: GPT-3.5, ChatGPT, GPT-4, and GPT-5 architecture details are not public

ChatGPT

- Built upon GPT-3.5, fine-tuned for conversational usage
- Joint use of supervised learning and reinforcement learning
- Reinforcement Learning from Human Feedback (RLHF)
 - 1st step: supervised fine-tuning
 - 2nd step: reinforcement learning

GPT-4

- **Multimodal** large language model: text & **images** as input
- Trained using both publicly available data and “data licensed from third-party providers”
- Fine-tuned using RLHF (as in ChatGPT)
- Context window: from 2,048 (GPT-3) and 4,096 (GPT-3.5) to 8,192 and 32,768 tokens (around 52 pages of text)

GPT-4

- Further details are not publicly available

“Given both the competitive landscape and the safety implications of large-scale models like GPT-4, this report contains no further details about the architecture (including model size), hardware, training compute, dataset construction, training method, or similar.”

GPT-4 – Example of visual input

User What is funny about this image? Describe it panel by panel.



Source: <https://www.reddit.com/r/hmmm/comments/ubab5v/hmmm/>

GPT-4 The image shows a package for a "Lightning Cable" adapter with three panels.

Panel 1: A smartphone with a VGA connector (a large, blue, 15-pin connector typically used for computer monitors) plugged into its charging port.

Panel 2: The package for the "Lightning Cable" adapter with a picture of a VGA connector on it.

Panel 3: A close-up of the VGA connector with a small Lightning connector (used for charging iPhones and other Apple devices) at the end.

The humor in this image comes from the absurdity of plugging a large, outdated VGA connector into a small, modern smartphone charging port.

GPT-4 – Example of visual input

User Answer question I.1.a. Think step-by-step.

I. Principe de la détection de rayonnement avec un bolomètre

Comme illustré sur la figure 1, un bolomètre est constitué d'un absorbeur qui reçoit le rayonnement que l'on désire détecter. Sa température T , supposée uniforme, est mesurée à l'aide d'un thermomètre incorporé, constitué d'un matériau conducteur dont la résistance $R(T)$ varie avec la température T ; cette variation est caractérisée par le coefficient $\alpha = \frac{1}{R} \frac{dR}{dT}$. L'ensemble possède la capacité thermique C_{th} .

Un barreau, conducteur thermique, homogène, de longueur L , de section S et de conductivité thermique λ et sans échanges thermiques latéraux, relie le bolomètre à un thermostat de température T_b fixe.

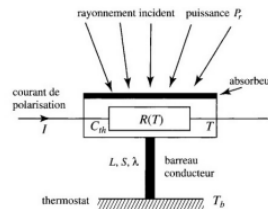


Figure 1 - Schéma d'un bolomètre

I.1. Dans cette première partie, on étudie le comportement du bolomètre en régime stationnaire, à la température de fonctionnement $T = T_b$.

I.1.a) Exprimer la température $T(x)$ en tout point du barreau conducteur repéré par son abscisse x ; on prendra $T(0) = T_0$ et $T(L) = T_b$.

GPT-4 To answer question I.1.a, we need to express the temperature $T(x)$ in every point of the conductive bar, which is characterized by the abscissa x . We are given that $T(0) = T_0$ and $T(L) = T_b$.

For a stationary regime, the heat equation for the one-dimensional heat conduction in the bar is:

$$d^2T/dx^2 = 0$$

Integrating once, we get:

$$dT/dx = C_1$$

Integrating a second time:

$$T(x) = C_1x + C_2$$

To determine the constants C_1 and C_2 , we use the boundary conditions:

$$T(0) = T_0, \text{ which gives } C_2 = T_0$$

$$T(L) = T_b, \text{ which gives } C_1 = (T_b - T_0)/L$$

Finally, the expression for $T(x)$ is:

$$T(x) = (T_b - T_0) * (x/L) + T_0$$

This represents a linear temperature profile along the conductive bar, with a slope of $(T_b - T_0)/L$.

Possible applications of LLMs

- Content Generation
- Summarization
- Machine Translation
- Personalized Recommendations
- Sentiment Analysis
- Coding Assistance
- Healthcare Support

Example 1 – Content Generation

Compose an email introducing a new product and its key features to potential customers.

Subject: Introducing Our Exciting New Product - [Product Name]

Dear [Customer's Name],

I hope this email finds you well. I am delighted to share some exciting news with you today. We are thrilled to introduce our latest product innovation, [Product Name], designed to revolutionize the way you [specific problem it solves or benefit it offers].

At [Your Company], we have always been committed to delivering cutting-edge solutions that enhance our customers' lives. With [Product Name], we have taken a significant leap forward in empowering you to [describe the main purpose or benefit of the product]. It combines advanced technology and thoughtful design to provide you with a seamless and extraordinary experience.

Example 1 – Content Generation

Here are some key features that make [Product Name] stand out:

1. [Feature 1]: [Describe the first feature and how it adds value to the customers]
 - [Highlight its unique qualities or benefits]
2. [Feature 2]: [Explain the second feature and how it addresses specific customer needs]
 - [Illustrate how this feature enhances the overall user experience]
3. [Feature 3]: [Describe the third feature and its impact on customer productivity or convenience]
 - [Emphasize how this feature sets [Product Name] apart from competitors]

Thank you for your time, and we look forward to having you as one of our valued customers, enjoying the extraordinary benefits of [Product Name].

Best regards,

[Your Name]

[Your Title/Position]

[Your Company Name]

Example 2 – Machine Translation

Translate the following sentence to Italian, Spanish, Greek, Arabic and Chinese: "Large Language Models are very powerful tools that are revolutionizing our daily lives."
Then, rewrite it using English slang.

Italian: "I modelli di lingua di grandi dimensioni sono strumenti molto potenti che stanno rivoluzionando la nostra vita quotidiana."

Spanish: "Los Modelos de Lenguaje Grandes son herramientas muy poderosas que están revolucionando nuestras vidas diarias."

Greek: "Τα μεγάλα μοντέλα γλωσσικής ανάπτυξης είναι πολύ ισχυρά εργαλεία που επανασχεδιάζουν την καθημερινή μας ζωή."

Arabic: "النماذج اللغوية الكبيرة هي أدوات قوية جدًا تحدث ثورة في حياتنا اليومية."

Chinese: "大型语言模型是非常强大的工具，正在彻底改变我们的日常生活。"

English slang: "Big Language Models are hella powerful tools that are totally flipping our everyday lives, man."

Example 3 – Sentiment Analysis

Assess the sentiment of this customer review: 'The customer service was terrible, and the product quality was disappointing.'

Based on the provided customer review, the sentiment expressed is negative. The customer mentions that the customer service was terrible and expresses disappointment with the product quality. These negative descriptors indicate an overall negative sentiment towards the customer's experience with both the customer service and the product.

Example 4 – Healthcare Support

Based on the following symptoms, analyze and provide possible diagnoses for a 50-year-old male patient: severe headache localized on the right side, sensitivity to light and sound, blurred vision.

Based on the symptoms provided, several potential diagnoses can be considered. However, please note that I am an AI language model and not a healthcare professional. It's important to consult with a qualified healthcare provider for an accurate diagnosis. Here are some possible diagnoses:

1. **Migraine:** Migraines often present as severe headaches that are localized on one side of the head, accompanied by sensitivity to light and sound. Blurred vision can also be experienced during a migraine attack.
2. **Cluster headache:** Cluster headaches typically cause severe, sharp pain on one side of the head, often around the eye area. Sensitivity to light and sound can accompany cluster headaches, and in some cases, blurred vision may also occur.
3. **Tension headache:** Tension headaches can cause moderate to severe head pain that may be localized on one side. While sensitivity to light and sound can be associated with tension headaches, blurred vision is less common.

Limitations & Failure cases of LLMs

- Lack of common sense and contextual understanding
 - Nonsensical responses when reasoning is needed (hallucination)
 - Rely on patterns learned from training data
- Propagation of biases and stereotypes
 - Reflect or amplify biases in training data
 - If not monitored, can reinforce social inequalities
- Limited knowledge and outdated information
 - Lack of knowledge beyond training data
 - Lack awareness of recent events
- Ethical, privacy, and security issues
 - Need for comprehensive rules for trustworthy AI

■ <https://www.europarl.europa.eu/news/en/press-room/20231206IPR15699/artificial-intelligence-act-deal-on-comprehensive-rules-for-trustworthy-ai#:~:text=The%20AI%20Act%20sets%20rules,of%20technology%20by%20public%20authorities.>

Example of hallucination

- Playing with GPT-3...

Get started ×

Enter an instruction or select a preset, and watch the API respond with a **completion** that attempts to match the context or pattern you provided.

You can control which **model** completes your request by changing the model.

KEEP IN MIND

- 🚩 Use good judgment when sharing outputs, and attribute them to your name or company. [Learn more.](#)
- 📄 Requests submitted to our API will not be used to train or improve future models. [Learn more.](#)
- 📅 Our default models' training data cuts off in 2021, so they may not have knowledge of current events.

Playground

Load a preset... ▾ Save

Does mRNA vaccines modify the DNA?

No, mRNA vaccines do not modify the DNA. Instead, they deliver instructions to cells on how to make a harmless protein that triggers an immune response to the virus, without ever entering the nucleus of the cell where the DNA is located.

<https://platform.openai.com/playground>

Example of hallucination

- Fact checking

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Example of hallucination

- Deepening the search...

Get started ×

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Playground

Load a preset... ▾ Save

What is the Kraken SARS-CoV-2 variant?

The Kraken SARS-CoV-2 variant is a new strain of the virus that first emerged in Denmark in January 2021. It was identified as a result of whole-genome sequencing of the virus, which revealed the presence of several mutations that could potentially make the virus more contagious than the original strain. The Kraken SARS-CoV-2 variant is believed to be more transmissible than the original strain, but it is not yet known if it is more likely to cause severe symptoms or lead to more hospitalizations and deaths.

<https://platform.openai.com/playground>

Example of hallucination

- Fact checking

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<https://platform.openai.com/playground>

Environmental cost of LLMs

- Training and running LLMs require significant hardware infrastructures, including data center and cooling systems
- This leads to high energy consumption and carbon footprint, contributing to climate change
- Research efforts are underway to develop more energy-efficient training strategies
- Example: BLOOM required 3.5 months and consumed 1,082,990 compute hours. Training was conducted on 48 nodes, each having 8 NVIDIA A100 80GB GPUs

Definition of “agent”

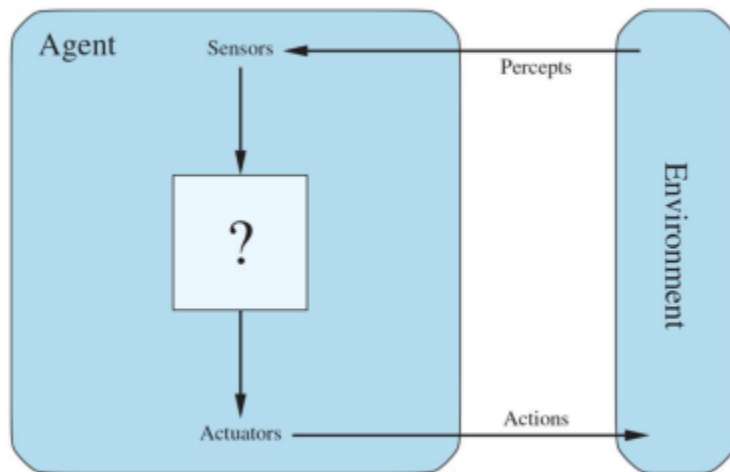
- AI model capable of reasoning, planning, and interacting with its environment
- System that **leverages an AI model to interact with its environment** to achieve a **user-defined objective**
- To fulfill the task, it combines
 - reasoning
 - planning
 - execution of actions (often via external tools)

Agents: basic idea



<https://huggingface.co/learn/agents-course/unit1/what-are-agents>

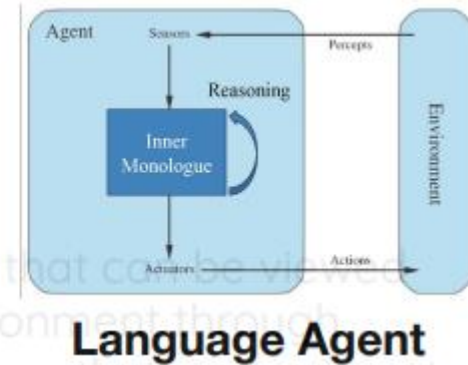
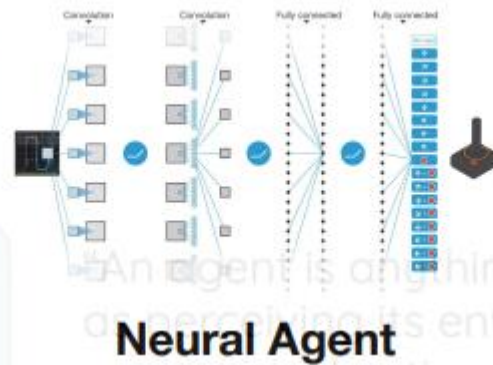
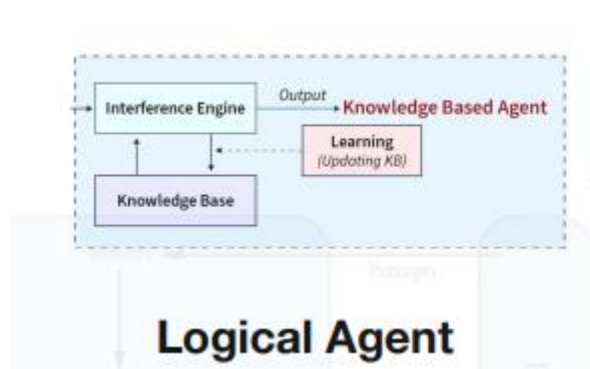
Alternative definition



“An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.”

-- Russell & Norvig, *AI: A Modern Approach* (2020)

History of agents



Logical Agent

Neural Agent

Language Agent

	Logical Agent	Neural Agent	Language Agent
Expressiveness	Low bounded by the logical language	Medium anything a (small) NN can encode	High almost anything, esp. verbalizable parts of the world
Reasoning	Logical inferences sound, explicit, rigid	Parametric inferences stochastic, implicit, rigid	Language-based inferences fuzzy, semi-explicit, flexible
Adaptivity	Low bounded by knowledge curation	Medium data-driven but sample inefficient	High strong prior from LLMs + language use

Mnih et al., "Human-level control through deep reinforcement learning." Nature (2015)

LLM-based agent

- The LLM serves as the main brain, agent module, or coordinator of the system
- It is activated using a prompt template that entails important details about how the agent will operate, and the tools it will have access to (along with tool details)
- While not mandatory, an agent can be profiled or be assigned a persona to define its role. This profiling information is typically written in the prompt

Agents and LLMs... Why?

- AI agents integrated with LLM(s) can use language as a vehicle for reasoning and communication
 - Instruction following, in-context learning, output customization reasoning (for better acting): state inferences, self-reflection, replanning, etc.

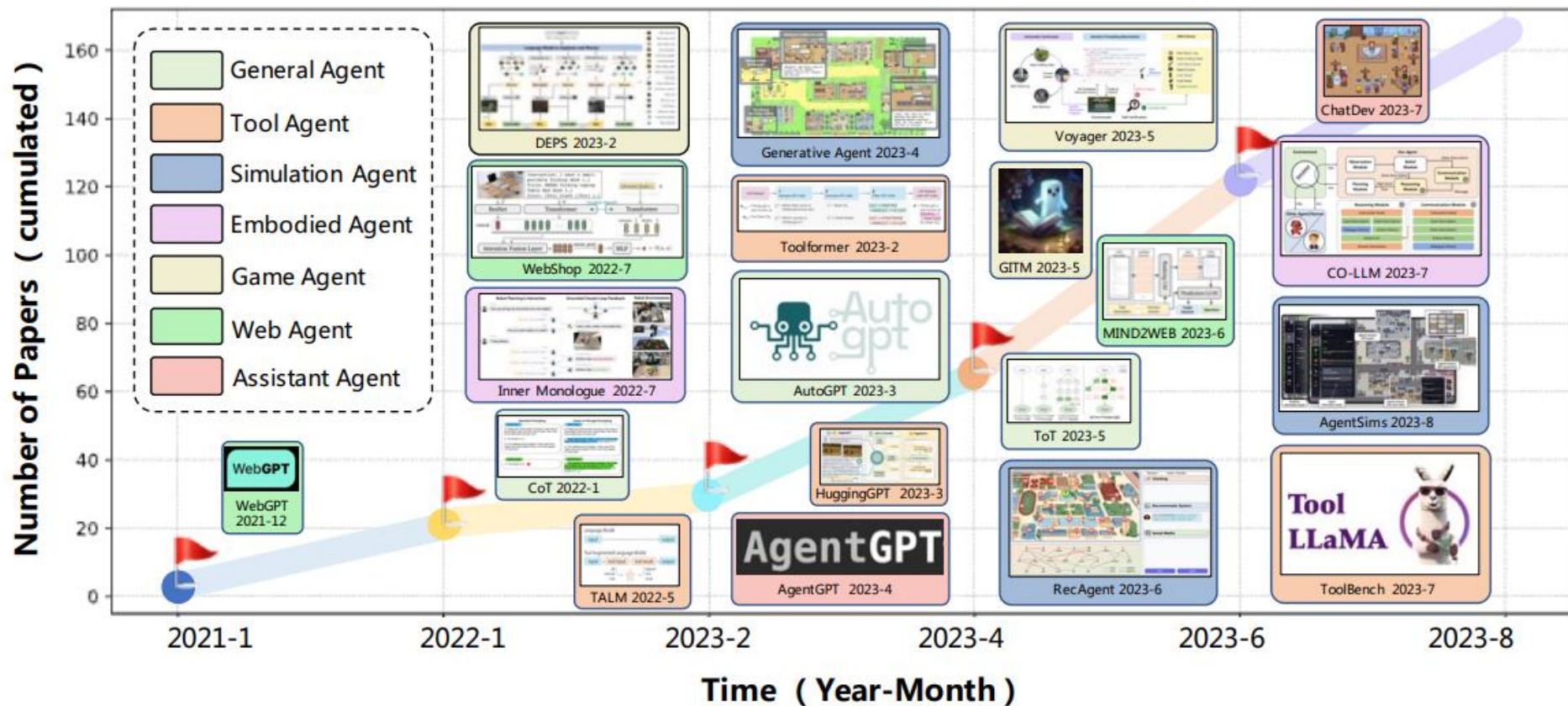
LLM agent frameworks

- LLM applications that can execute complex tasks through the use of an architecture that combines LLMs with key modules like planning and memory
- The LLM serves as the main controller or "brain" that controls a flow of operations needed to complete a task or user request
- Key elements
 - **Brain:** AI model that handles reasoning and planning
 - **Body:** Everything the agent is equipped to do

Examples of applications

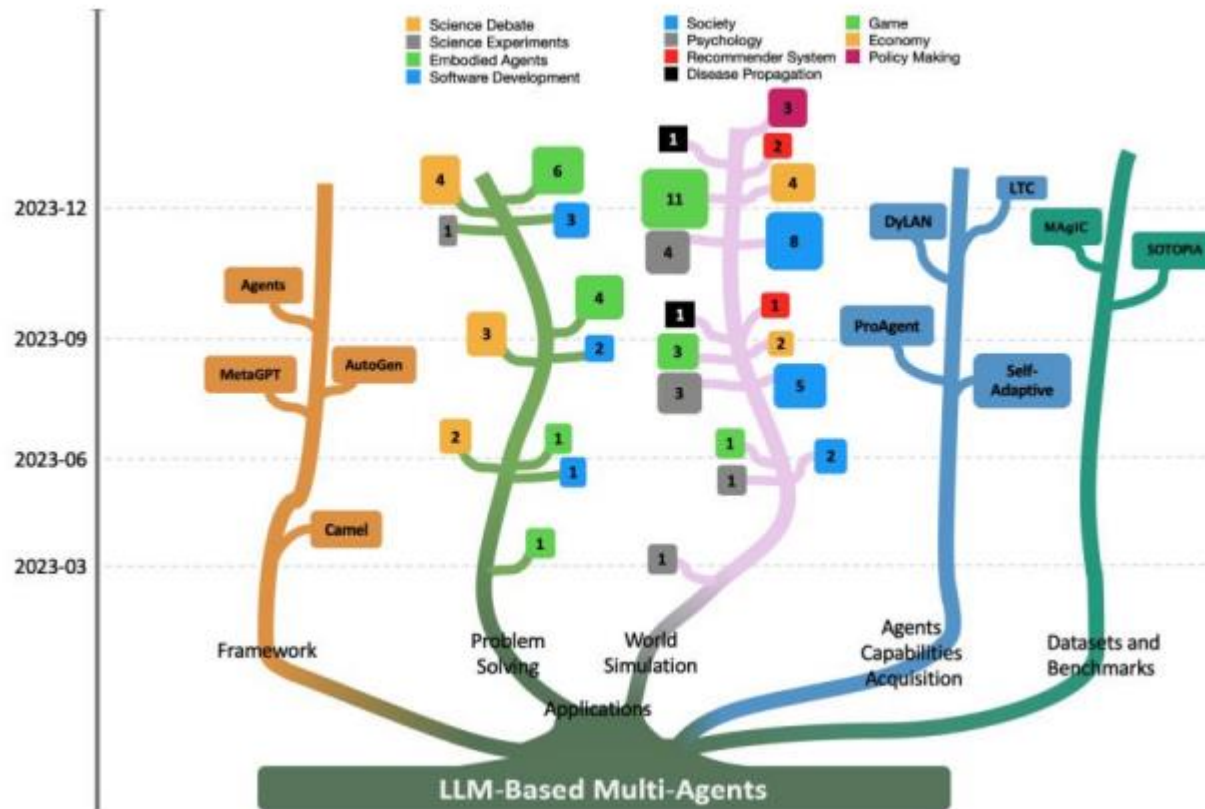
- Virtual assistant
- Customer service chatbot
- Penetration tester
- Coding
- Gaming
- ...

LLM agent frameworks



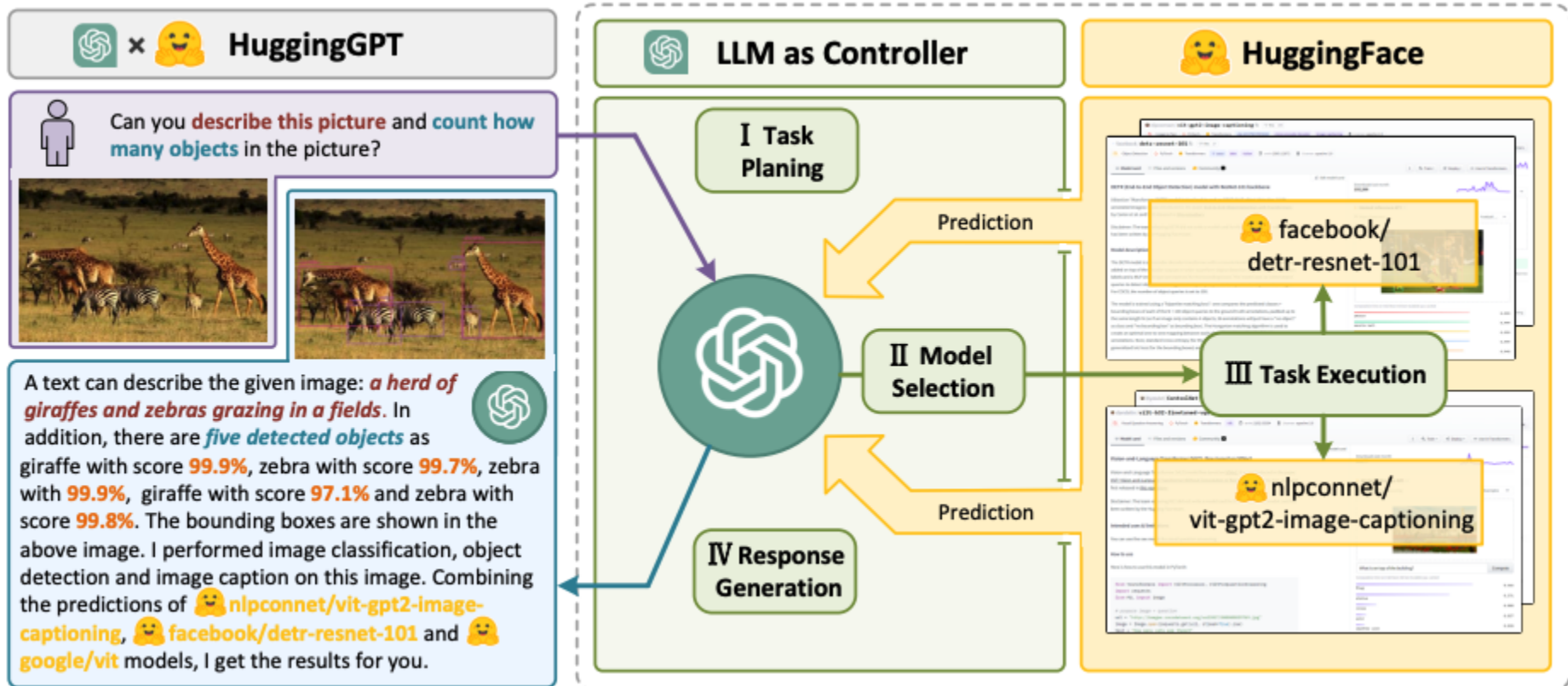
<https://arxiv.org/pdf/2308.11432>

LLM agents' domains of application



Guo, Taicheng, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V. Chawla, Olaf Wiest, and Xiangliang Zhang. "Large language model based multi-agents: A survey of progress and challenges." arXiv preprint arXiv:2402.01680 (2024).

Examples of tools' applications



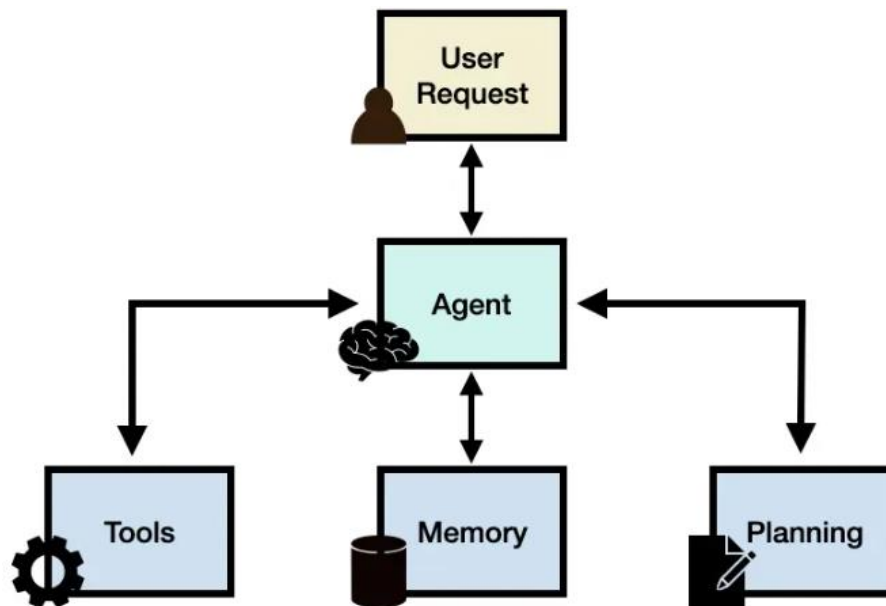
<https://arxiv.org/pdf/2308.11432>

Agents: typical features

- **Understand natural language:** Interpret and respond to human instructions in a meaningful way.
- **Reason and plan:** Analyze information, make decisions, and devise strategies to solve problems.
- **Interact with its environment:** Gather information, take actions, and observe the results of those actions.

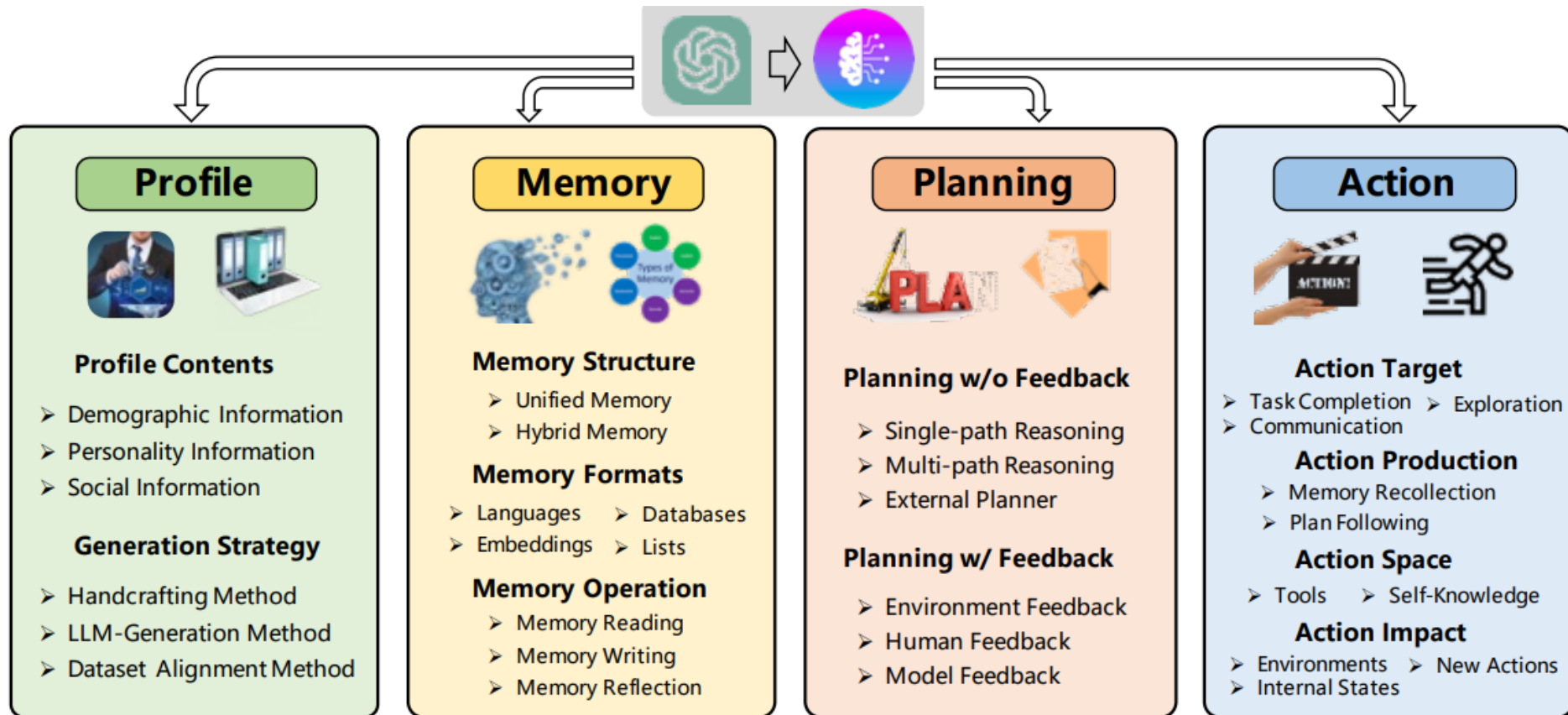
LLM agent frameworks

- Core components of a LLM agent
 - **User Request:** a user question or request
 - **Agent/Brain:** the agent core acting as coordinator
 - **Planning:** assists the agent in planning future actions
 - **Memory:** manages the agent's past behaviors



<https://arxiv.org/pdf/2308.11432>

LLM agent frameworks



<https://arxiv.org/pdf/2308.11432>

Profiling module

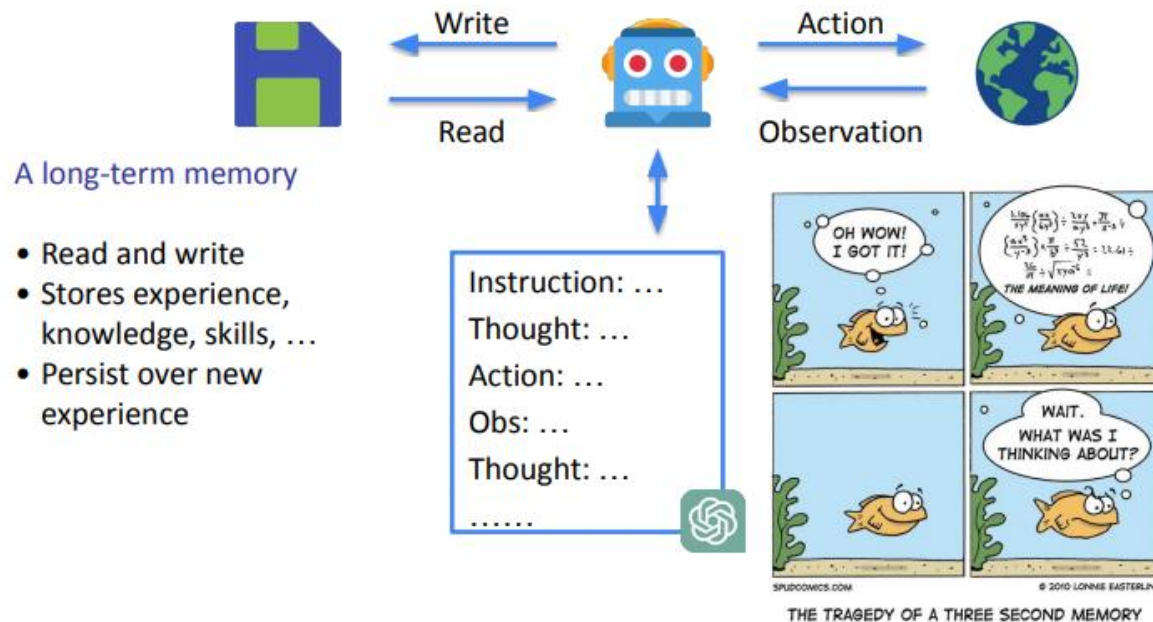
- Autonomous agents typically perform tasks by assuming specific roles, such as coders, teachers, and domain experts
- The profiling module aims to indicate the profiles of the agent roles, which are usually written into the prompt to influence the LLM behaviors
- Agent profiles typically encompass basic information such as age, gender, and career, as well as psychology information, reflecting the personalities of the agent, and social information, detailing the relationships between agents

1) Profiling module

- Profiling strategies
 - **Handcrafted**: manually specified in the prompt
 - **LLM-generated**: automatically generated based on the seed information
 - **Data-driven**: aligned to real-world datasets

2) Memory module

- Language agents interact with external environments and internal memories (information-storing devices)
 - Interact with **short-term memory** (context window): reasoning
 - Interact with **long-term memory** (LLM weights, event logs, codebase, prompt library, etc.): retrieving and learning

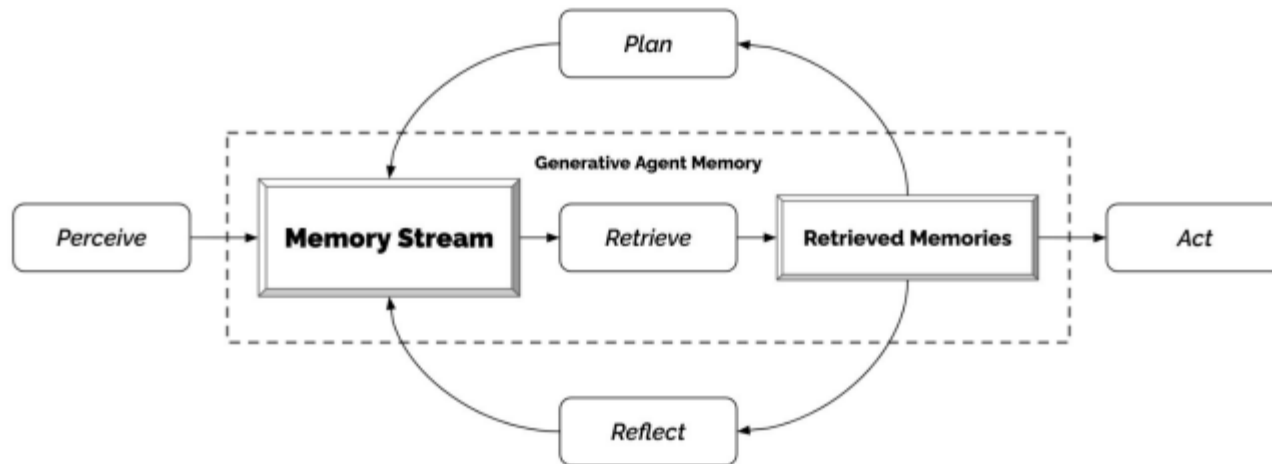


<https://language-agent-tutorial.github.io/>

2) Memory module

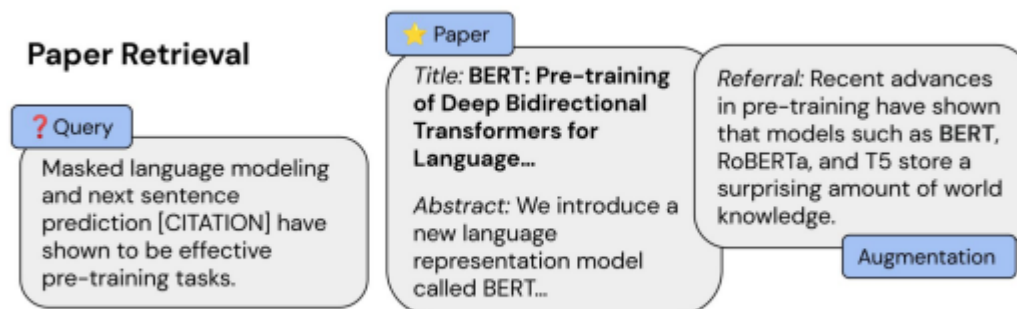
- It helps to store the agent's internal logs including past thoughts, actions, and observations from the environment, including all interactions between agent and user.
- Memory types
 - **Short-term memory**
 - It includes context information about the agent's current situations
 - It is typically realized by in-context learning which means
 - It is short and finite due to context window constraints
 - **Long-term memory**
 - It includes the agent's past behaviors and thoughts that need to be retained and recalled over an extended period of time
 - It often leverages an external vector store accessible through fast and scalable retrieval to provide relevant information for the agent as needed

2) Memory module



2) Referral augmentation

- Use reasoning to augment what it is stored in memory



Tang et al. "Referral Augmentation for Zero-Shot Information Retrieval. "

Reasoning

- **For humans:** various mental processes
- **For LMs:** intermediate generation
- **For agents:** internal actions



Reasoning without acting: example

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

Standard

Answer: iPod

X

Reason Only

Thought: Let's think step by step. Apple Remote was originally designed to interact with Apple TV. Apple TV can be controlled by iPhone, iPad, and iPod Touch. So the answer is iPhone, iPad, and iPod Touch.

Answer: iPhone, iPad, iPod Touch

X

Reasoning with acting: example

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.

Act 1: `Search[Apple Remote]`

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the `Front Row media center` program(long text omitted)

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.

Act 2: `Search[Front Row]`

Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]

Thought 3: Front Row is not found. I need to search `Front Row (software)` .

Act 3: `Search[Front Row (software)]`

Obs 3: Front Row is a discontinued media center software (long text omitted)

Thought 4: `Front Row (software)` is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.

Act 4: `Finish[keyboard function keys]`



Acting without reasoning: example

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

Act Only

Act 1: `Search`[Apple Remote]

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the Front Row media center program ...

Act 2: `Search`[Front Row]

Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]

Act 3: `Search`[Front Row (software)]

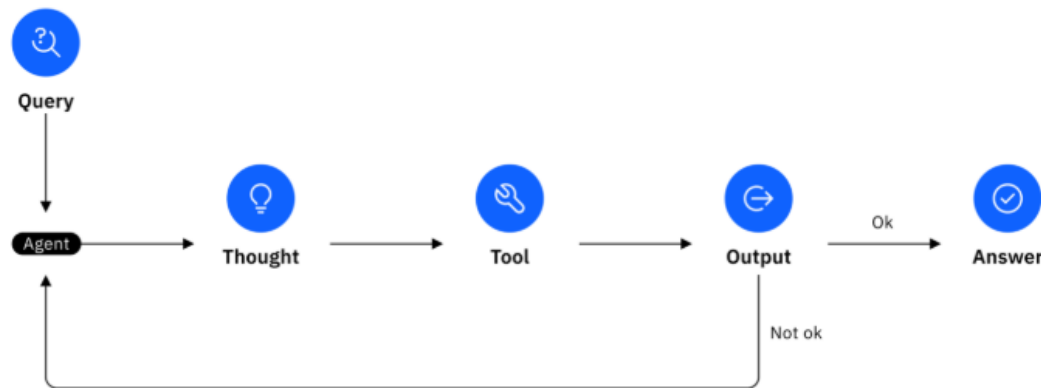
Obs 3: Front Row is a discontinued media center software ...

Act 4: `Finish`[yes]

X

The ReAct framework

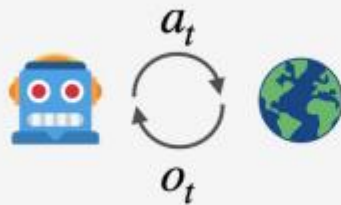
- ReAct can be understood most generally as a machine learning (ML) paradigm to integrate the reasoning and action-taking capabilities of LLMs.
- It inherently creates a feedback loop in which the model problem-solves by iteratively repeating this interleaved *thought-action-observation* process.
- Each time this loop is completed—that is, each time the agent has taken an action and made an observation based on the results of that action—the agent must then decide whether to repeat or end the loop.



Yao and others in the 2023 paper, “ReACT: Synergizing Reasoning and Acting in Language Models,”

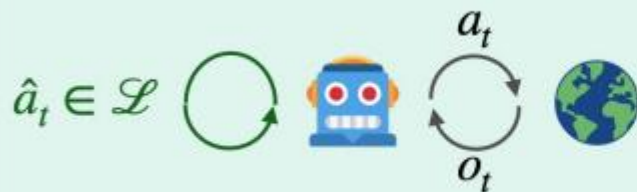
Traditional vs. ReAct spaces

- In traditional agents the action space is defined by the environment



- **External feedback** o_t
- Agent context $c_t = (o_1, a_1, o_2, a_2, \dots, o_t)$
- Agent action $a_t \sim \pi(a | c_t) \in A$

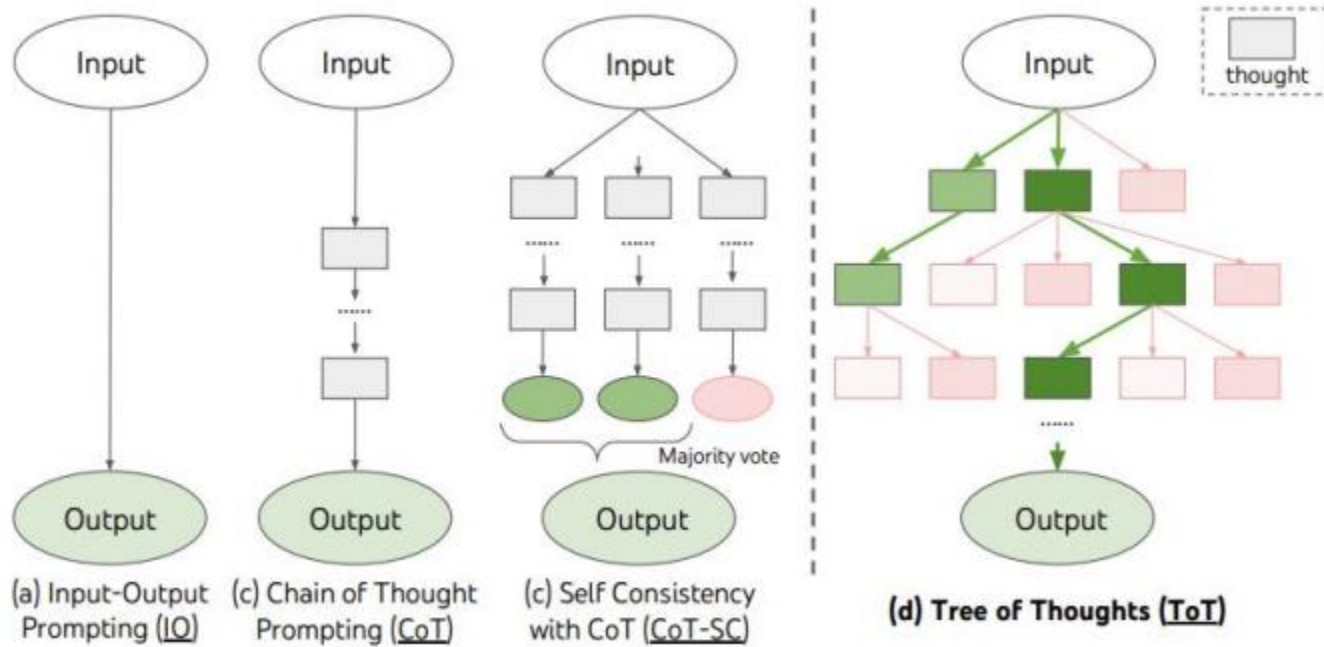
- In the ReAct agents (Reasoning+Acting) it is **augmented by reasoning**



- $\hat{a}_t \in \mathcal{L}$ can be any language sequence
- Agent context $c_{t+1} = (c_t, \hat{a}_t, a_t, o_{t+1})$
- $\hat{a}_t \in \mathcal{L}$ only updates **internal context**

<https://language-agent-tutorial.github.io/>

Tree of Thoughts



Planning

- The world model

A computational model of environment transition dynamics

$$\hat{T}: \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$$

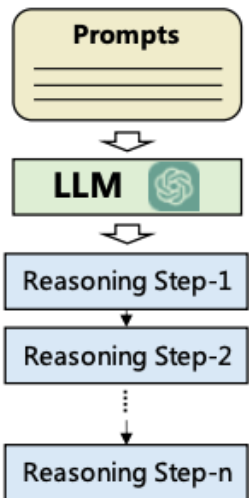
If I do this (a_t) right now (s_t), what would happen next (s_{t+1})?

Planning module without feedback

- It helps to break down the necessary steps or subtasks the agent will solve individually to answer the user request
- This step is important to enable the agent to reason better about the problem and reliably find a solution.
- The planning module will leverage an LLM to decompose a detailed plan which will include subtasks to help address the user question
- Commonly based on LLM in-context learning

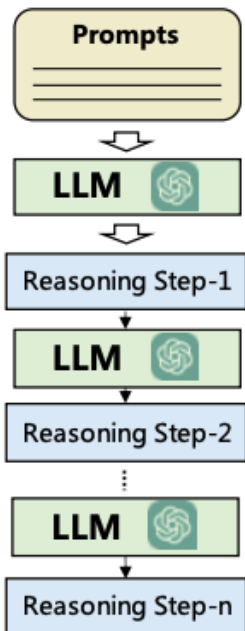
Planning module without feedback

CoT , Zero-shot Cot

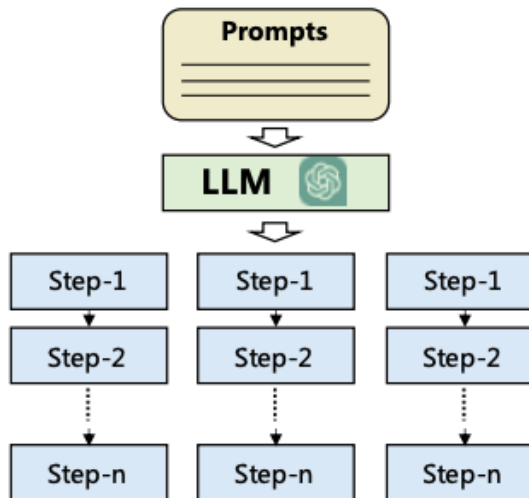


Single-Path Reasoning

ReWOO , HuggingGPT

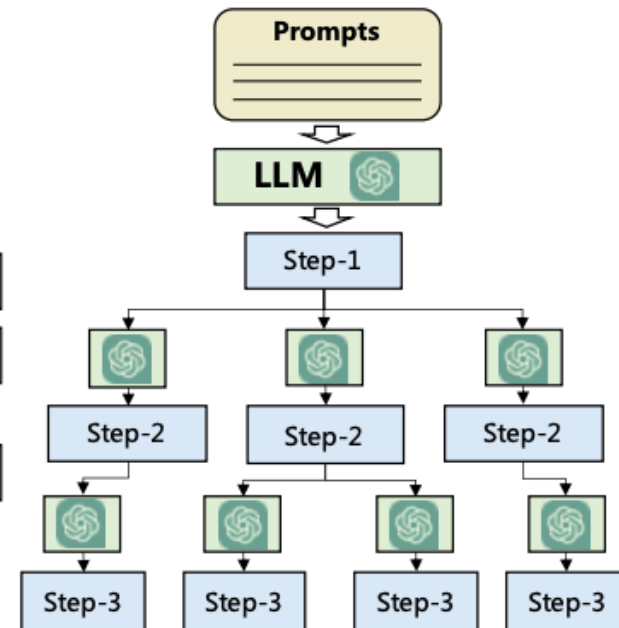


CoT-SC



Multi-Path Reasoning

ToT , LMZSP , RAP



Planning module with feedback

- The model iteratively reflects and refines the execution plan based on past actions and observations.
- The goal is to correct and improve on past mistakes which helps to improve the quality of final results.
- This is particularly important in complex real-world environments and tasks where trial and error are key to completing tasks.

Planning module with feedback

(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.

Act 1: `Search[Apple Remote]`

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the `Front Row media center` program ...

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Obs 2: Could not find `[Front Row]`. Similar: `['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]`

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Act 3: `Search[Front Row (software)]`

Obs 3: `Front Row` is a discontinued media center software ...

Thought 4: `Front Row (software)` is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.

Act 4: `Finish[keyboard function keys]`



Tools

- They enable the LLM agent to interact with external environments such as Wikipedia Search API, Code Interpreter, and Math Engine.
- They could also include databases, knowledge bases, and external models.
- When the agent interacts with external tools it executes tasks via workflows that assist the agent to obtain observations or necessary information to complete subtasks and satisfy the user request

Agent evaluation

- Three possible strategies
 - via benchmarks
 - via LLMs/VLMs
 - via crowdsourcing
- Challenges in agent evaluation
- Real-world environmental setup complexity
- Task coverage
 - Open-ended success criteria
 - Multiple valid solution paths
 - Cannot script evaluation metrics, need for human judgment
 - Evaluation beyond task success

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- Acknowledgements

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- Affiliation

- The author and his staff are currently members of the Database and Data Mining Group at Dipartimento di Automatica e Informatica (Politecnico di Torino) and of the SmartData interdepartmental centre
 - <https://dbdmg.polito.it>
 - <https://smartdata.polito.it>

Thank you!